

Sentinel Analyst Lab

Microsoft Sentinel Analyst Lab — Step-by-Step Guide

This manual lab guides you through creating a reusable Microsoft Sentinel analyst lab in the Azure portal. It covers data onboarding, analytics rules, incidents & investigations, Case Management, playbooks/automation, hunting & KQL, threat intelligence/watchlists, notebooks, and cleanup.

Estimated time: 2–4 hours. Assumes Owner/Contributor on an Azure subscription. Default region used in examples: East US — change to your preferred region.

Prerequisites

1. Azure subscription and Owner/Contributor on the subscription or a resource group.
2. Azure AD account able to grant admin consent for connectors (Office 365, Teams, ServiceNow connectors require admin consent).
3. Optional: two small VMs (Windows + Linux) for agent data (B1s/B2s recommended).
4. Familiarity with Azure Portal and PowerShell / Bash.

What you'll create and test

1. Resource group, Log Analytics workspace, Microsoft Sentinel enabled.
2. Data connectors: Azure Activity, Azure AD Sign-in, VM agents (Azure Monitor Agent), optional Office 365/CEF.
3. Generate test data (Data Collector API, logger, Windows event creation).
4. Create scheduled analytics rules (template + custom RDP brute-force rule).
5. Trigger incidents, triage, investigate (entity pages, graph).
6. Enable and use Cases (native Case Management).

7. Create and run a Logic Apps playbook and attach it to an analytics rule.
 8. Create hunting queries, bookmarks, saved KQL.
 9. Upload threat intelligence watchlist and reference it.
 10. Open and run a Sentinel notebook (MSTICPy sample).
 11. Cleanup resources and manage costs.
-

A. Create core resources

1. Create Resource Group

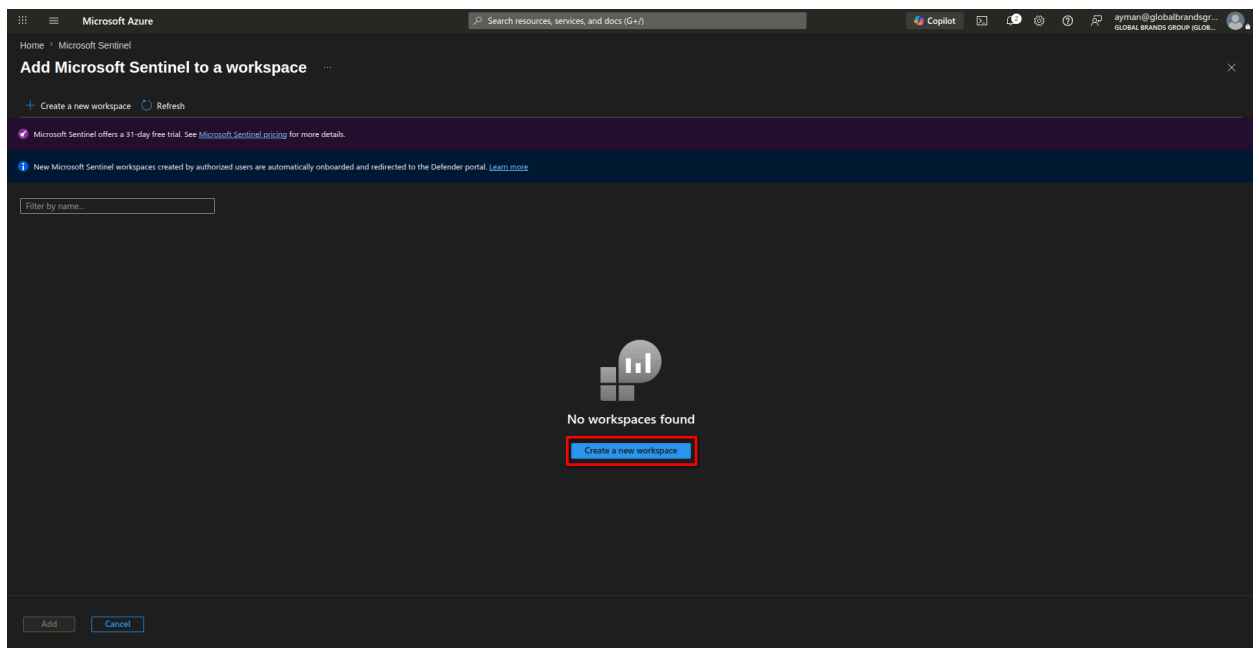
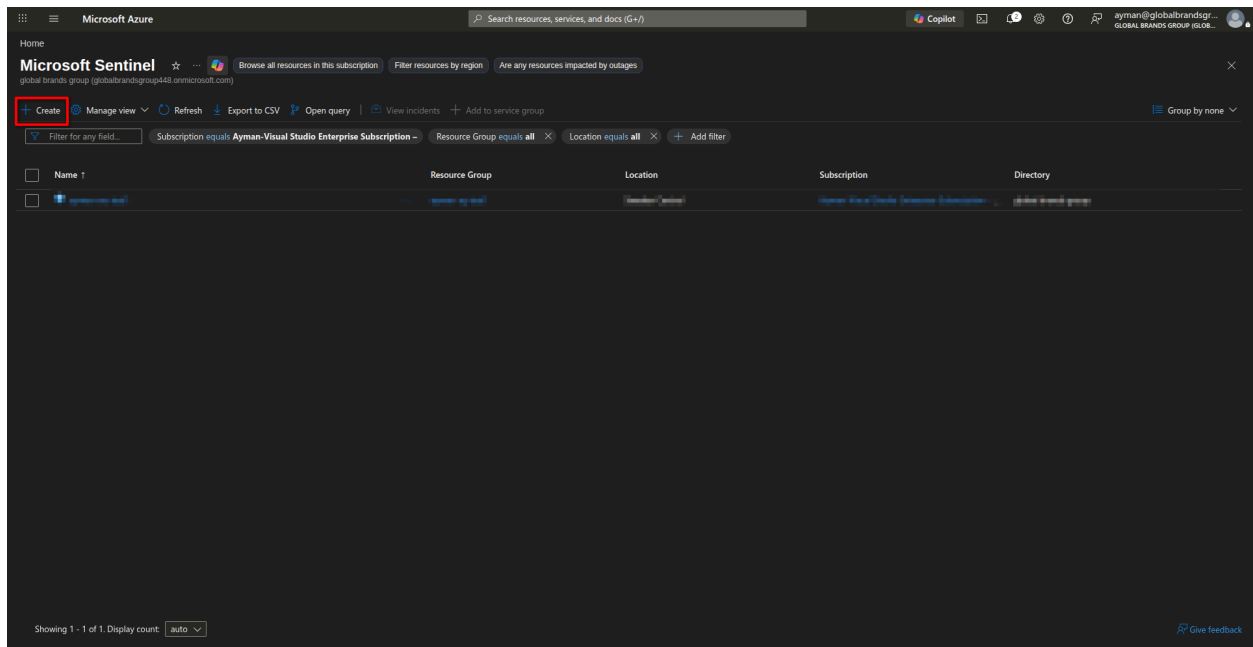
- Portal: Home → Resource groups → + Create
- Choose Subscription and Region (East US in examples)
- Name: `Sentinel-Lab-rg` → Review + Create

2. Create Log Analytics workspace

- Portal: Home → Log Analytics workspaces → + Create
- Select Resource Group `Sentinel-Lab-rg`, Region, Name: `sentinellab-law`
- Pricing: Pay-as-you-go. Retention: 7–30 days to limit cost → Create

3. Enable Microsoft Sentinel

- Portal: Home → Microsoft Sentinel → + Add → select `sentinellab-law` → Add



Microsoft Azure

Home > Microsoft Sentinel > Add Microsoft Sentinel to a workspace

Create Log Analytics workspace

Basics Tags Review & Create

A Log Analytics workspace is the basic management unit of Azure Monitor Logs. There are specific considerations you should take when creating a new Log Analytics workspace. [Learn more](#)

With Azure Monitor Logs you can easily store, retain, and query data collected from your monitored resources in Azure and other environments for valuable insights. A Log Analytics workspace is the logical storage unit where your log data is collected and stored.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription: Ayman-Visual Studio Enterprise Subscription - MPN

Resource group: (New) Sentinel-Lab-rg

[Create new](#)

Instance details

Name: sentinel-lab

Region: Sweden Central

[Review & Create](#) [Previous](#) [Next: Tags >](#)

Microsoft Azure

Home > Microsoft Sentinel

Add Microsoft Sentinel to a workspace

[+ Create a new workspace](#) [Refresh](#)

Microsoft Sentinel offers a 31-day free trial. See [Microsoft Sentinel pricing](#) for more details.

New Microsoft Sentinel workspaces created by authorized users are automatically onboarded and redirected to the Defender portal. [Learn more](#)

Filter by name:

Workspace	Location	ResourceGroup	Subscription	Directory
sentinel-lab	swedencentral	sentinel-lab-rg	Ayman-Visual Studio Enterprise Subscription - MPN	global brands group

[Add](#) [Cancel](#)

B. Onboard data sources (connectors)

1. Windows & Linux VMs (Azure Monitor Agent)

- Create small VM if needed: Portal → Virtual machines → + Create Windows Server 2019

Microsoft Azure

Home > Compute infrastructure > Virtual machines

Create a virtual machine

Basics Disks Networking Management Monitoring Advanced Tags Review + create

This is a preview of a new Create-VM experience. Features are limited in this experience and all feedback is appreciated. [Click here to access the previous experience.](#)

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription: Ayman-Visual Studio Enterprise Subscription – MPN

Resource group: Sentinel-Lab-rg

[Create new](#)

Instance details

Virtual machine name: winServer2019

Region: (Europe) Sweden Central

[Deploy to an Azure Extended Zone](#)

Availability options: Availability zone

Zone options: ☒ Self selected zone
Select a specific zone for your VM

☐ Azure-selected zone (Preview)
Let Azure assign the best zone for your needs

Availability zone: Zone 1

Previous Next **Review + create** [Give feedback](#)

Microsoft Azure

Home > Compute infrastructure > Virtual machines

Create a virtual machine

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Security type: Standard

Image: Windows Server 2022 Datacenter – x64 Gen2

[See all images](#) [Configure VM generation](#)

VM architecture: ☒ x64

Size: Standard_D2s_v3 – 2 vcpus, 8 GiB memory (\$74.46/month)

[See all sizes](#) [Customize cores](#)

Run with Azure Spot discount: ☐

Administrator account

Username: M0hammad

Password:

Confirm password:

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports: ☒ Allow selected ports

Select inbound ports: RDP (3389)

Previous Next **Review + create** [Give feedback](#)

2. Windows Security Events

- Sentinel → Content Hub → Windows Security Events → Install

Microsoft Defender

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Find Defender solutions

Home

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Search

Threat management

Content management

Content hub

Repositories

Community

Configuration

Tables

Data connectors

Analytics

Summary rules

Watchlist

Automation

Identities

Endpoints

Email & collaboration

Cloud apps

Cloud security

Didn't find what you were looking for? We're showing a limited set of results. Try refining your search for more specific results. [Learn more](#)

Search: windows security events

Status: All Content type: All Support: All Provider: All Category: All Content sources: All

Install/Update Delete

Content title	Status	Content source	Provider	Support
Windows Security Events	Not installed	Solution	Microsoft	Microsoft
Windows Security Events via AMA	Not installed	Solution	Microsoft	Microsoft
Security Events via Legacy Agent	Not installed	Solution	Microsoft	Microsoft
NRT Security Event log cleared	Not installed	Solution	Microsoft	Microsoft
Suspicious Windows Login Outside Normal Ho...	Not installed	Solution	Microsoft	Microsoft
Windows System Time changed on hosts	Not installed	Solution	Microsoft	Microsoft
Windows System Shutdown/Reboot(Sysmon)	Not installed	Solution	Microsoft	Microsoft
Excessive Windows Logon Failures	Not installed	Solution	Microsoft	Microsoft
NRT Base64 Encoded Windows Process Comm...	Not installed	Solution	Microsoft	Microsoft
New EXE deployed via Default Domain or Defa...	Not installed	Solution	Microsoft	Microsoft
Gain Code Execution on ADFS Server via SMB ...	Not installed	Solution	Microsoft	Microsoft
Starting or Stopping HealthService to Avoid De...	Not installed	Solution	Microsoft	Microsoft
Process Execution Frequency Anomaly	Not installed	Solution	Microsoft	Microsoft
AD FS Remote Auth Sync Connection	Not installed	Solution	Microsoft	Microsoft

Windows Security Events

Microsoft Provider 3.0.12 Version

Description

Note: Please refer to the following before installing the solution:

- Review the solution [Release Notes](#)
- There may be [known issues](#) pertaining to this Solution, please refer to them before installing.

The Windows Security Events solution for Microsoft Sentinel allows you to ingest Security events from your Windows machines using the Windows Agent into Microsoft Sentinel. This solution includes two (2) data connectors to help ingest the logs.

- Windows Security Events via AMA** - This data connector helps in ingesting Security Events logs into your Log Analytics Workspace using the new Azure Monitor Agent. [Learn more about ingesting using the new Azure Monitor Agent here](#). **Microsoft recommends using this Data Connector.**
- Security Events via Legacy Agent** - This data connector helps in ingesting Security Events logs into your Log Analytics Workspace using the legacy Log Analytics agent.

Data Connectors: 2 Workbooks: 2 Analytic Rules: 20 Hunting Queries: 50

[Learn more about Microsoft Sentinel](#) | [Learn more about Solutions](#)

Content type: 20 Analytics rule 2 Data connector 50 Hunting query

Category: Security - Threat Protection

Pricing: Free

[Install](#) [View details](#)

Microsoft Defender

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Threat intelligence

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Content hub

Repositories

Community

Configuration

Tables

Data connectors

Analytics

Summary rules

Watchlist

Automation

Identities

Endpoints

Email & collaboration

Cloud apps

Cloud security

Content hub > Windows Security Events > Data connectors

Refresh Guides Workspace: sentinellab-lab

Device specific AMA connectors have been deprecated. [Learn more](#)

In anticipation of the End of Support of the HTTP Data Collector API, Azure Function-based data connectors using legacy API will not be supported after September 14, 2026. Please update to CCF-based data connectors. [Learn more](#)

4 Connectors 2 Connected More content at Content Hub

Search by name or provider Providers: Microsoft

Data Types: All Status: All

Status	Connector name	Updates
Connected	Azure Activity Microsoft	...
Connected	Microsoft 365 Insider Ri... Microsoft	...
Deprecated	[DEPRECATED] Security ... Microsoft	...
Connected	Windows Security Event... Microsoft	...

Windows Security Events via AMA

Disconnected Status Microsoft Provider Last Log Received

Description

You can stream all security events from the Windows machines connected to your Microsoft Sentinel workspace using the Windows agent. This connection enables you to view dashboards, create custom alerts, and improve investigation. This gives you more insight into your organization's network and improves your security operation capabilities.

Last data received

Content source: Windows Security Events Version: 1.0.0

Author: Microsoft Supported by: Microsoft Corporation Email

Related content

0 Workbooks 1 Queries 20 Analytics rules templates

Data received: 4 3 [Go to log analytics](#)

[Open connector page](#)

Connector details

Windows Security Events via AMA

Disconnected Status Provider Last Log Received

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Last data received: --

Content source: Windows Security Events Version: 1.0.0

Author: Microsoft Supported by: Microsoft Corporation | Email

Related content: 0 Workbooks, 1 Queries, 20 Analytics rules templates

Data received: 0

Go to log analytics

Table management: Manage the table tiers - analytics and lake and table retention. Select the table to manage the table tier and configure its retention settings.

Table name	Tier	Table type	Analytics reten...	Total retention	Workspace
SecurityEvent	Analytics	Sentinel	30 days	30 days	sentinellab-law

Configuration: Enable data collection rule. Security Events logs are collected only from Windows agents.

Refresh

Rule name Created by Filter name

No results

+ Create data collection rule

Connector details

Windows Security Events via AMA

Disconnected Status Provider Last Log Received

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Configuration: Enable data collection rule. Security Events logs are collected only from Windows agents.

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Rule name Created by Filter name

No results

+ Create data collection rule

DCR Wizard

Basic Resources Collect Review + create

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all of your resources.

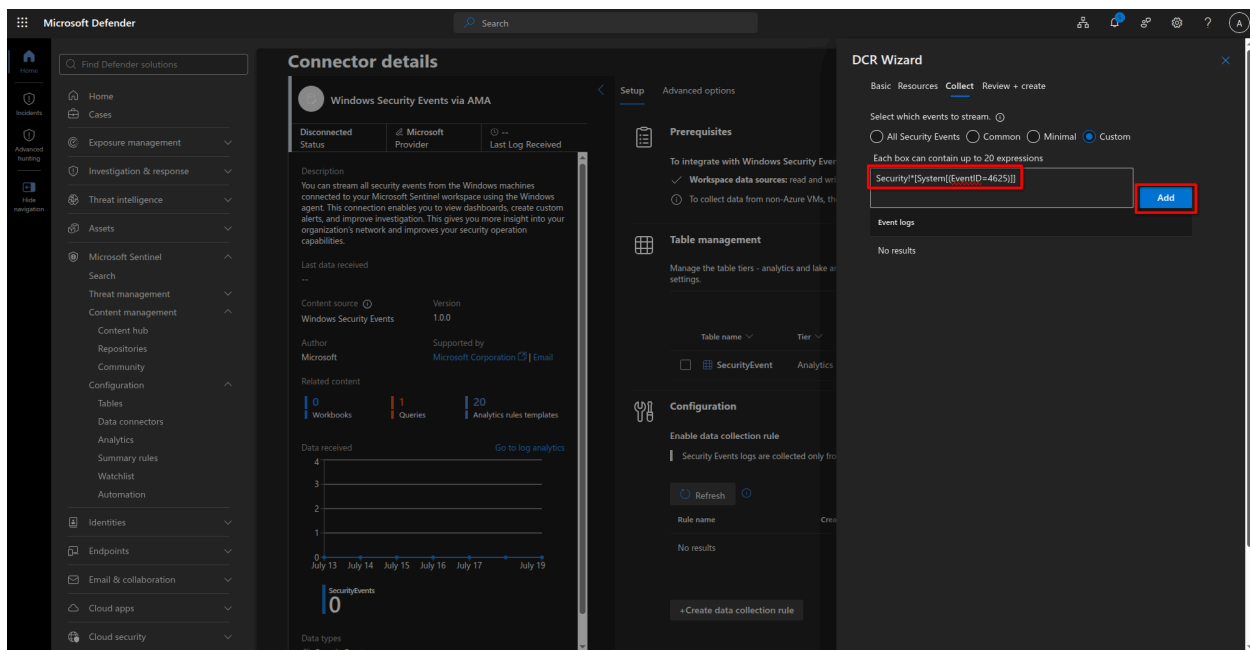
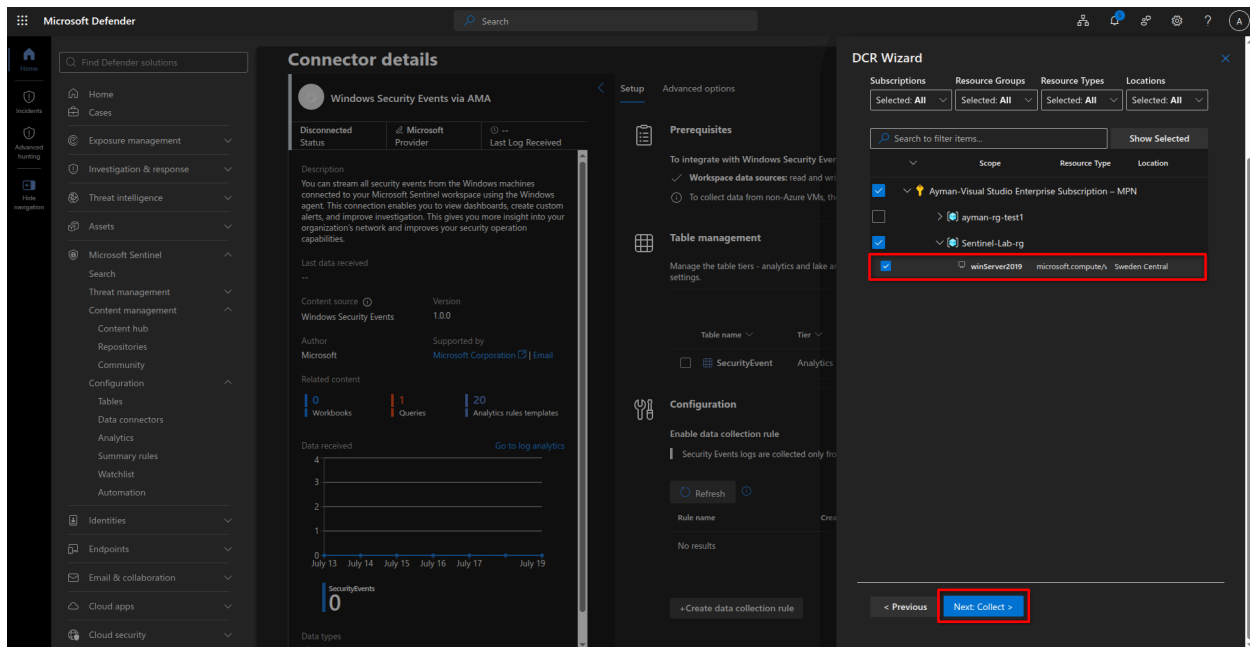
Rule details

Rule name: failedSignInLogs

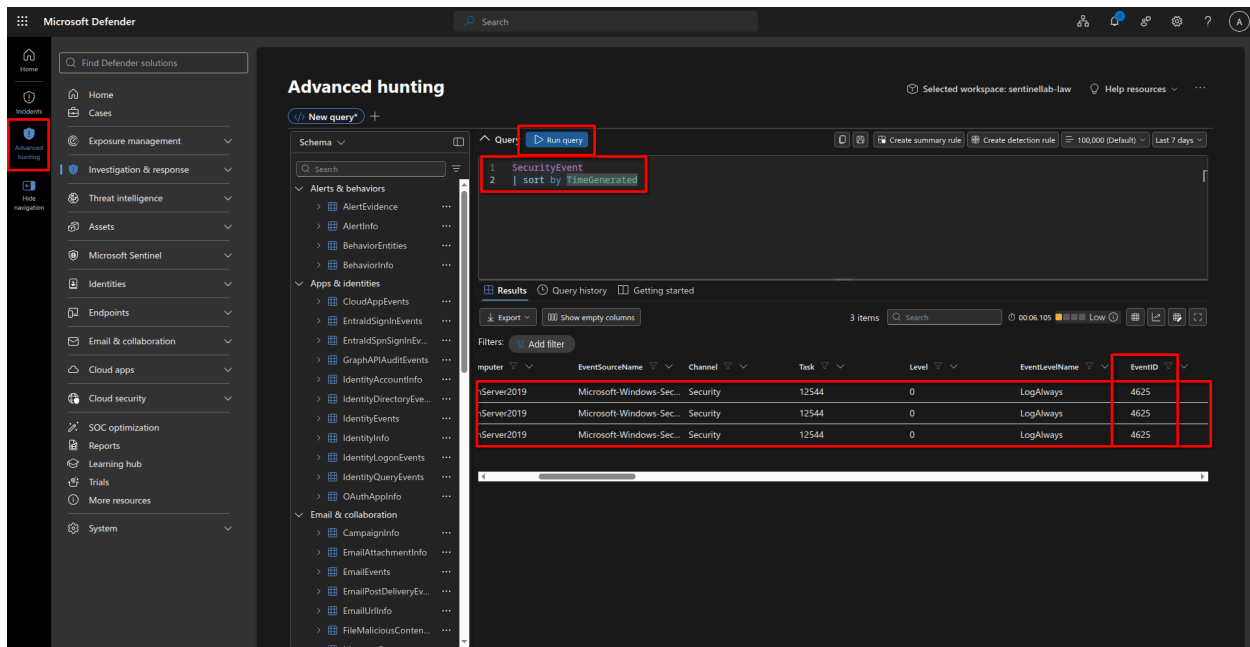
Subscription: Ayman-Visual Studio Enterprise Subsc...

Resource group: Sentinel-Lab-rg

Next: Resources



C. Test failed logins



D. Create analytics rules

1. Portal → Microsoft Sentinel → select your workspace.
2. Left menu → Configuration → Analytics → + Create → choose "Scheduled query rule".
4. In the Rule wizard:
 - Rule name: RDP Brute Force — Custom
 - Description: Detects repeated failed logons (RDP brute force) grouped by account/source/computer.
 - Severity: Medium (or adjust)
 - Tactics: MITRE ATT&CK — Credential Access

A. Prepare the query

- If you are using Windows Security events collected by the agent:
- Use this improved KQL (returns an alert row per (Account, SourceIp, Computer) that crosses threshold):

```
// RDP brute-force detection – Windows Security log (4625 = failed logon)
SecurityEvent
| where EventID == 4625
// adjust field names if your telemetry uses different columns (Account, TargetUserName, etc.)
| extend Account = tostring(Account), SourceIp = tostring(IpAddress), Computer = tostring(Computer)
| where isnotempty(Account)
| summarize attempts = count(), firstTime = min(TimeGenerated), lastTime = max(TimeGenerated) by Account, SourceIp, Computer
    | where attempts >= 3
| project TimeGenerated = lastTime, Account, SourceIp, Computer, attempts, firstTime, lastTime
```

5. Set Scheduling:

- Run query every: 5 minutes
- Lookup/Lookback period: 5 minutes (ensures recent events are considered)

6. Alert settings / Alert grouping:

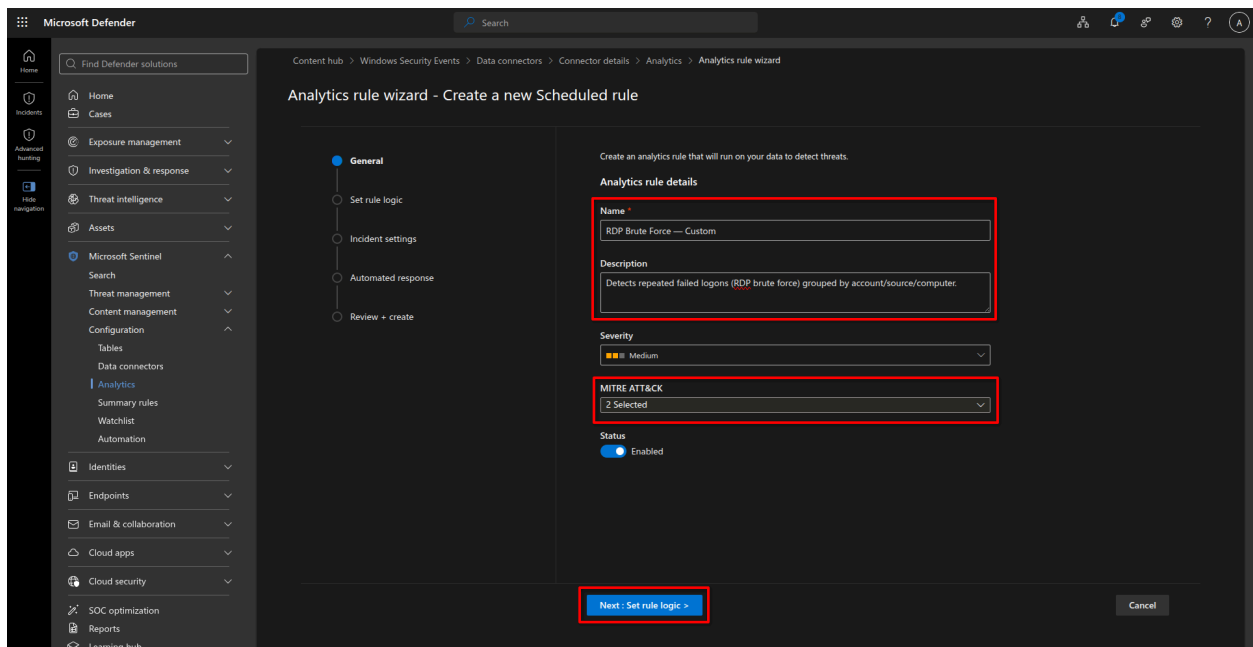
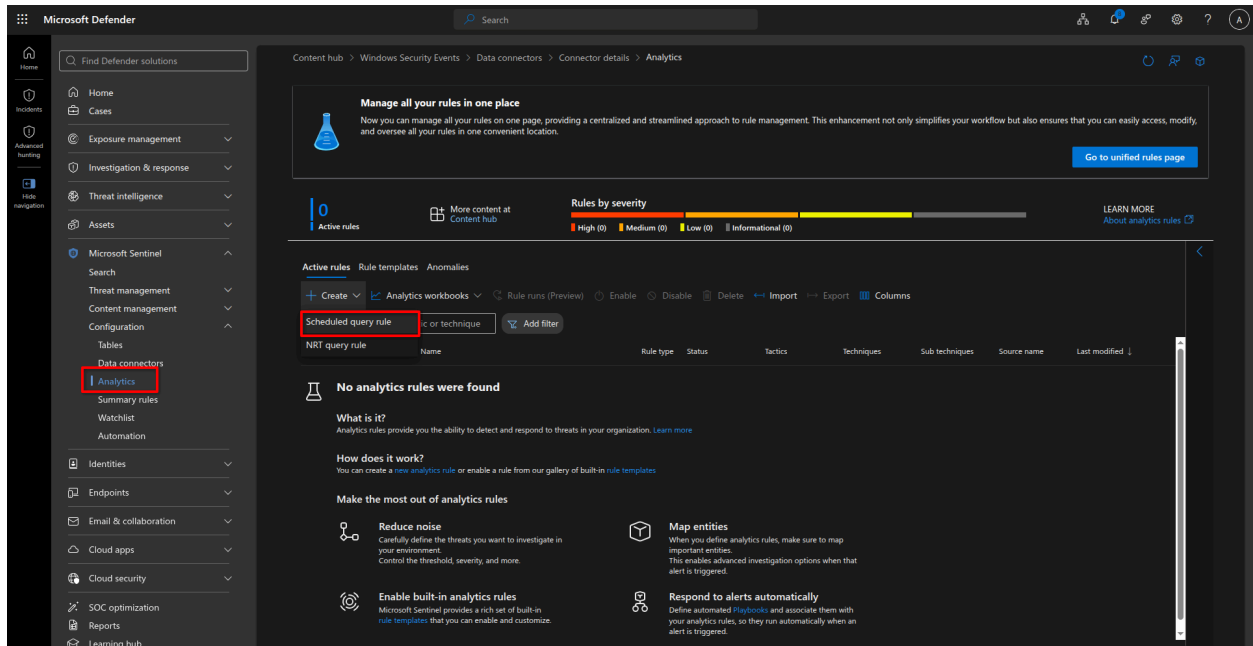
- Group alerts into incidents: choose to create incident per "Account" or leave default depending on preference.
- Grouping conditions: choose fields to group by — e.g., Account + SourceIp + Computer.

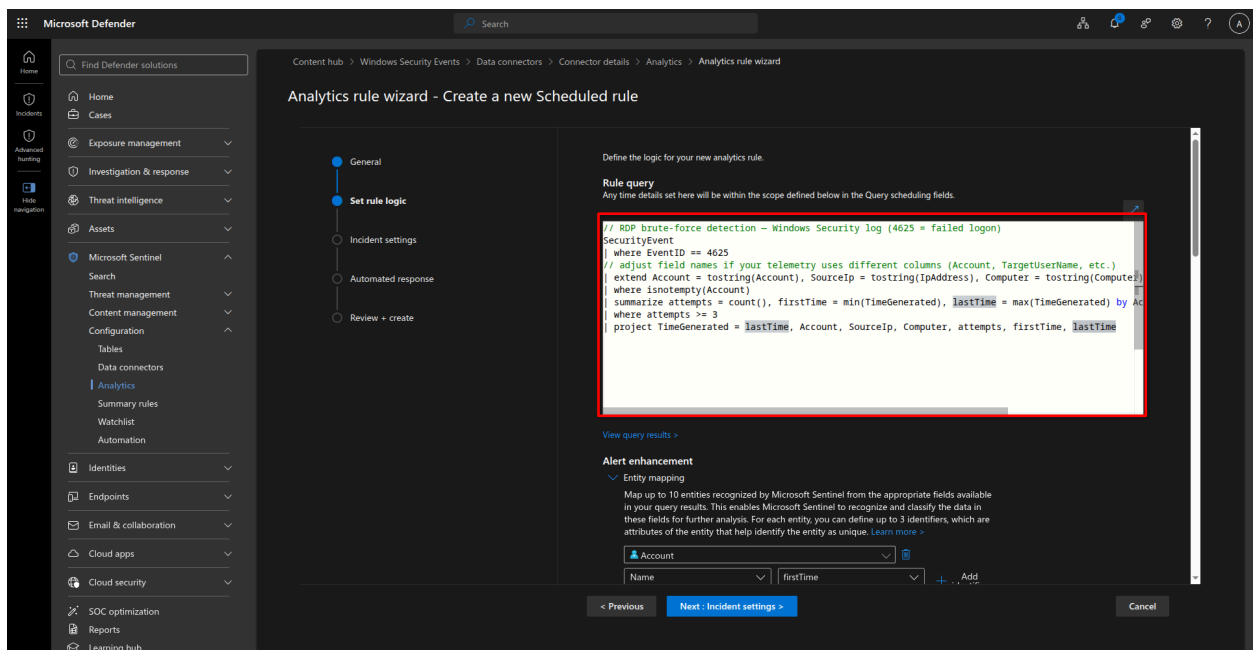
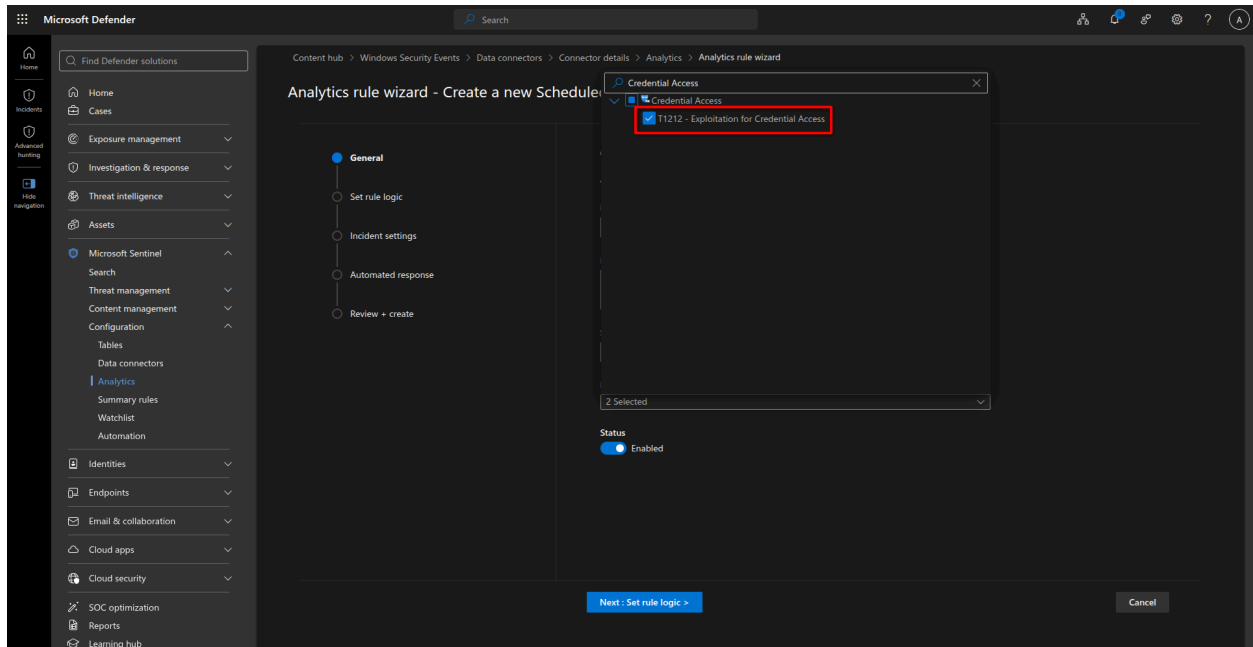
7. Set Incident Settings:

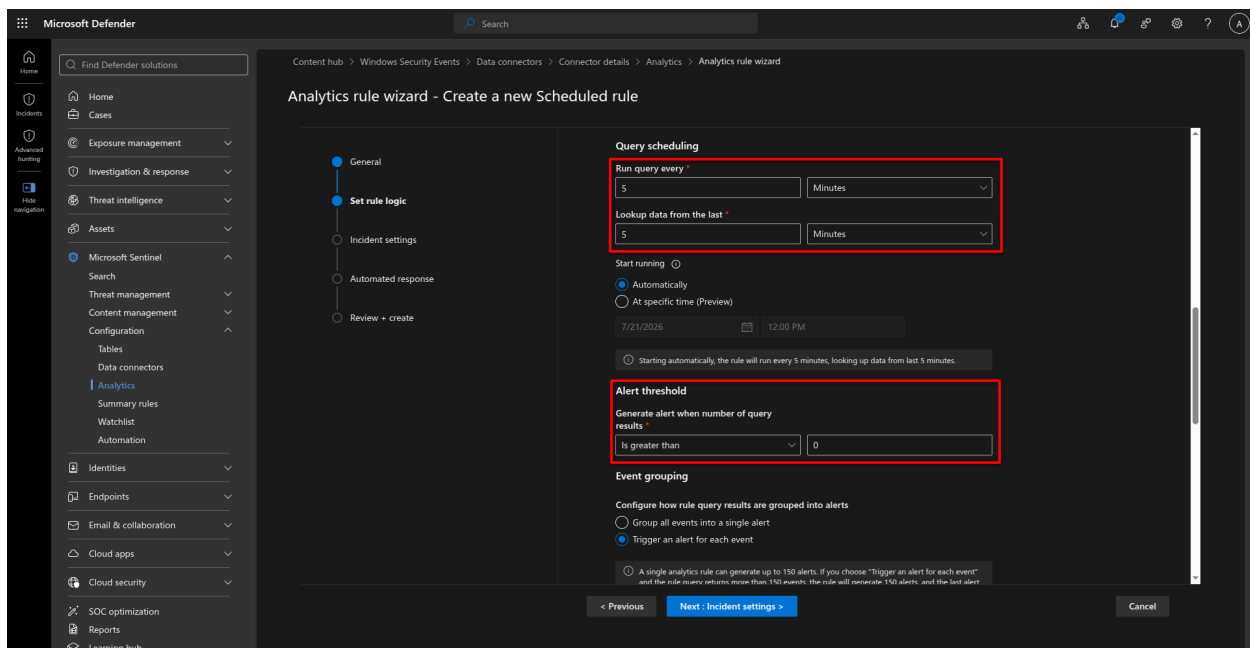
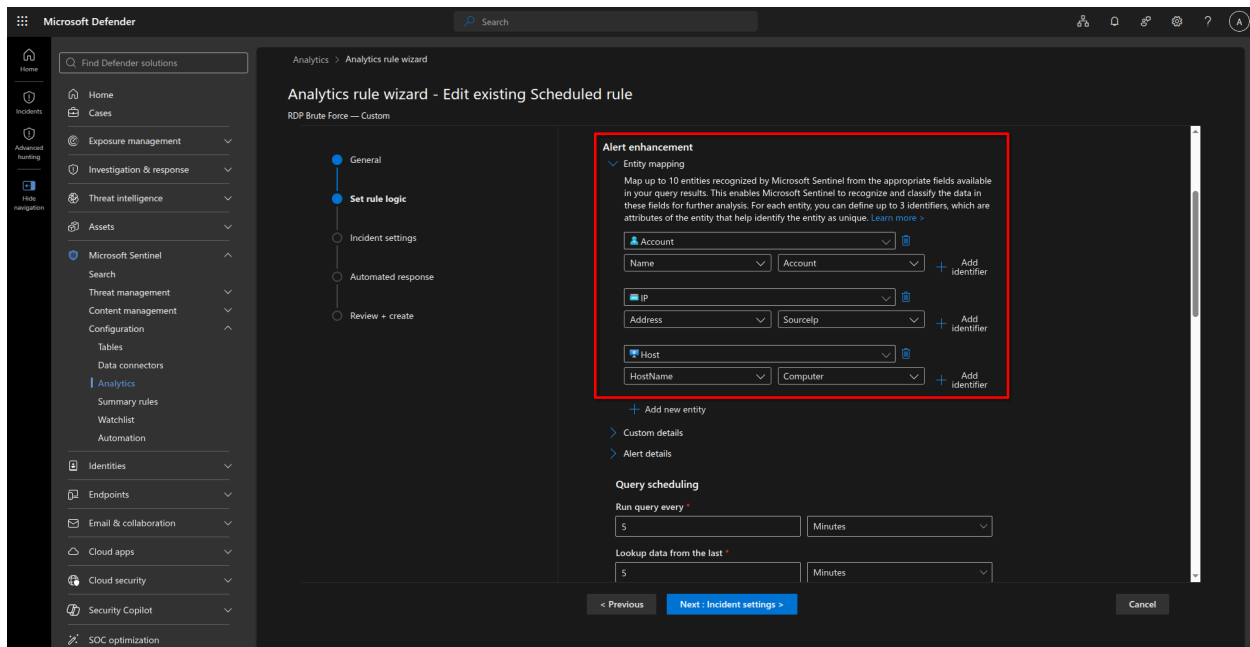
- Create incident when result count > 0
- Incident severity mapping: map to Medium.
- Add entities mapping: configure which query columns map to entity types:
 - Account → Account (User)

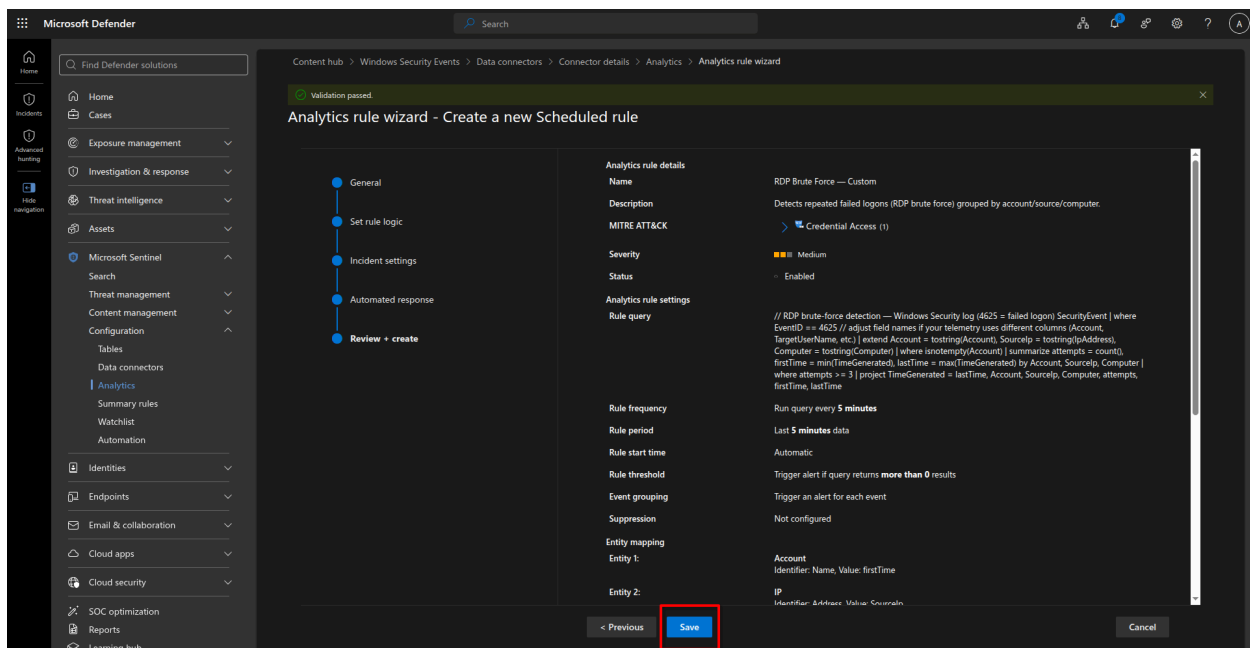
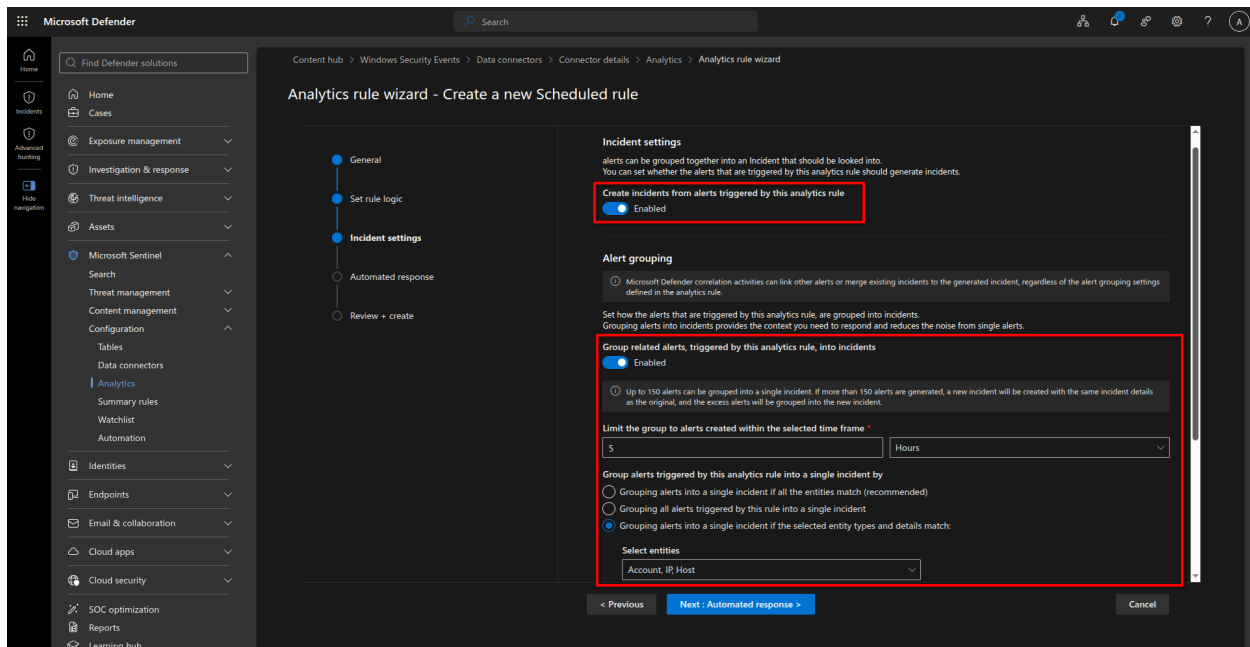
- Sourcelp → IP Address
- Computer → Hostname or Device

8. Review & Create.







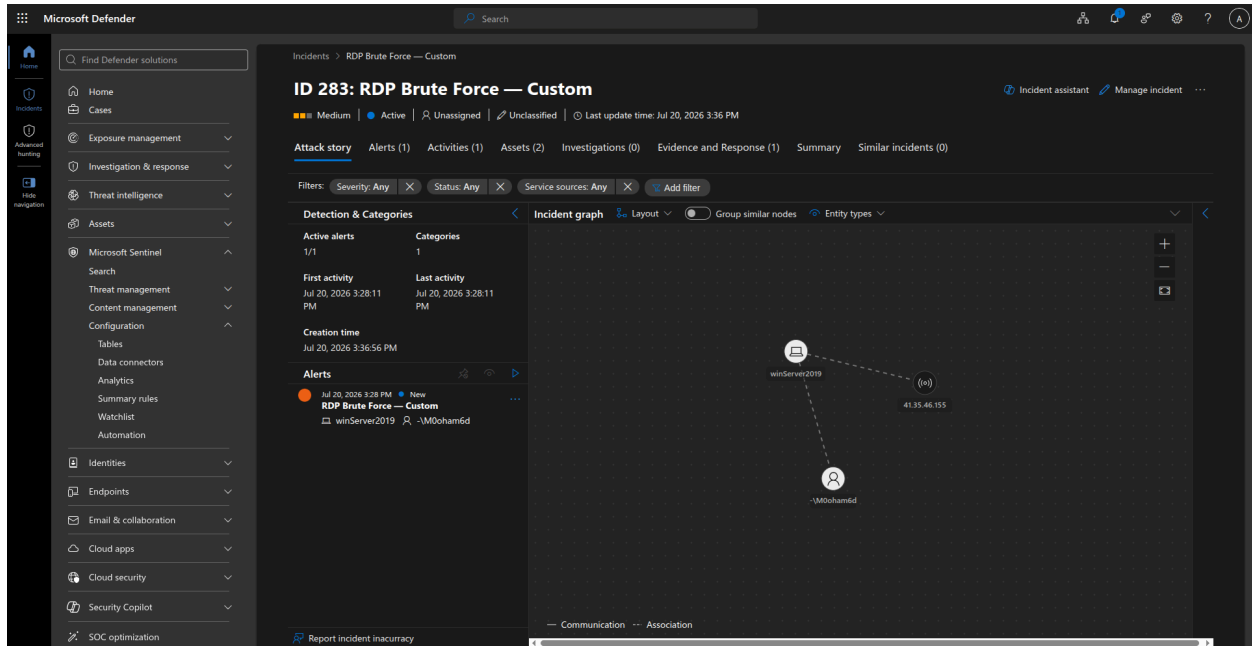


E. Incidents, triage, and Case Management

1. Trigger an incident

- Generate test events matching the analytics rule and wait for the scheduled rule to run

- Sentinel → Incidents → recent incidents
2. Triage an incident
 - Open incident → review Alerts, Entities, Timeline
 - Use the Investigation Graph tab to view relationships between entities (user, IP, device)

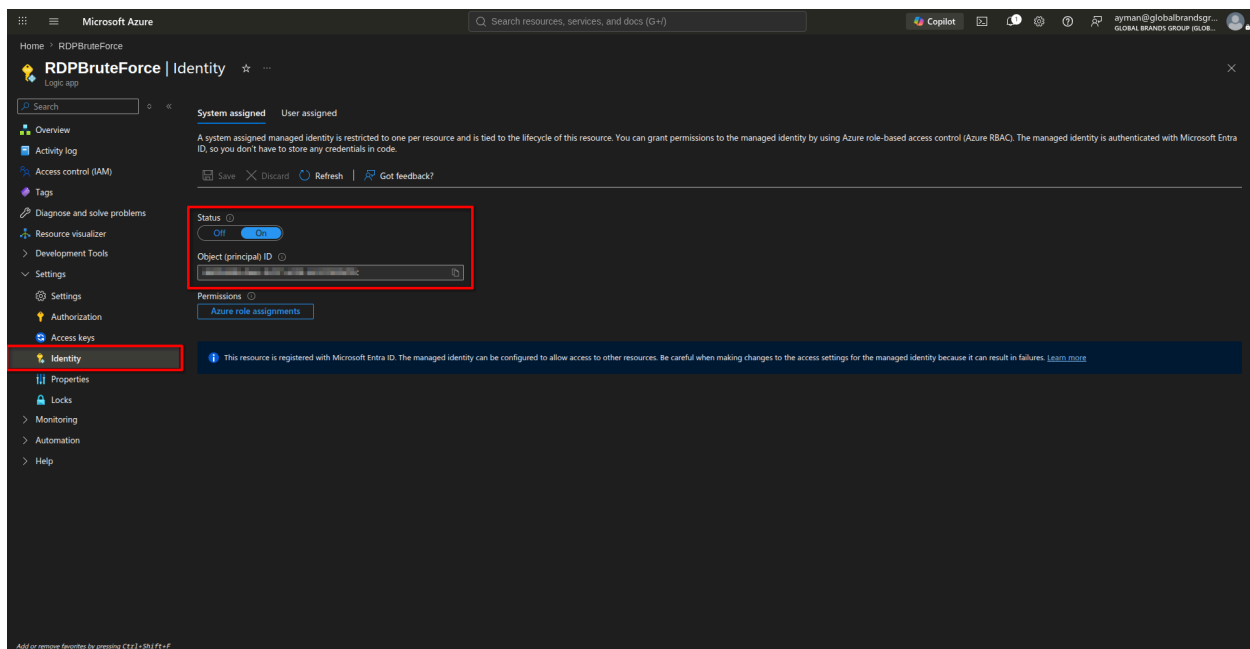


F. Automation / Playbooks (Logic Apps)

1. Enable the Managed Identity on the Logic App

Via Azure Portal:

1. Open your Logic App (playbook) → left menu → **Identity**
2. Under **System assigned** tab, toggle **Status** to **On**
3. Click **Save**
4. Azure generates an Object (principal) ID for the Logic App — copy this, you'll need it for role assignment



2. Assign RBAC Role on the Target VM (or Resource Group/Subscription)

The identity needs a role that includes

`Microsoft.Compute/virtualMachines/powerOff/action` and `deallocate/action`.

Recommended role: Virtual Machine Contributor (has power off/start/restart/deallocate rights, but not full VM management like disk/network deletion — better than Contributor for least privilege).

To assign it:

1. Go to the Resource Group → **Access control (IAM)**
2. **Add** → **Add role assignment**
3. Role: **Virtual Machine Contributor**
4. Assign access to: **Managed identity**
5. Select members → choose your Logic App's managed identity
6. **Review + assign**

Microsoft Azure

Home > Sentinel-Lab-rg | Access control (IAM)

Search resources, services, and docs (G+)

Overview Activity log Access control (IAM) Tags Resource visualizer Events Settings Cost Management Cost analysis Cost alerts (preview) Budgets Advisor recommendations Monitoring Insights (preview) Alerts Metrics Diagnostic settings Logs Advisor recommendations Workbooks Automation Help

Search

Add role assignment

Add custom role

Check access

Role assignments Roles Deny assignments

Looking for the previous check access view? [Click here.](#)

Check access

Review the level of access a user, group, service principal, or managed identity has to this resource. [Learn more](#)

ayman@globalbrandsgroup448.onmicrosoft.com Assignments

Active 2 Eligible 0 Deny 0

Search by role name or membership

Role Name	Scope	Membership	Condition	Action
Active permanent assignments (2)				
Foundry User	Subscription (Inherited)	ayman	No	-
Owner	Subscription (Inherited)	ayman	Yes	-
Active time-bound assignments (0)				
Eligible permanent assignments (0)				
Eligible time-bound assignments (0)				
Deny assignments (0)				

Microsoft Azure

Home > Sentinel-Lab-rg | Access control (IAM)

Add role assignment

Role Members Conditions Assignment type Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

[Copilot can help pick a role](#)

Job function roles Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

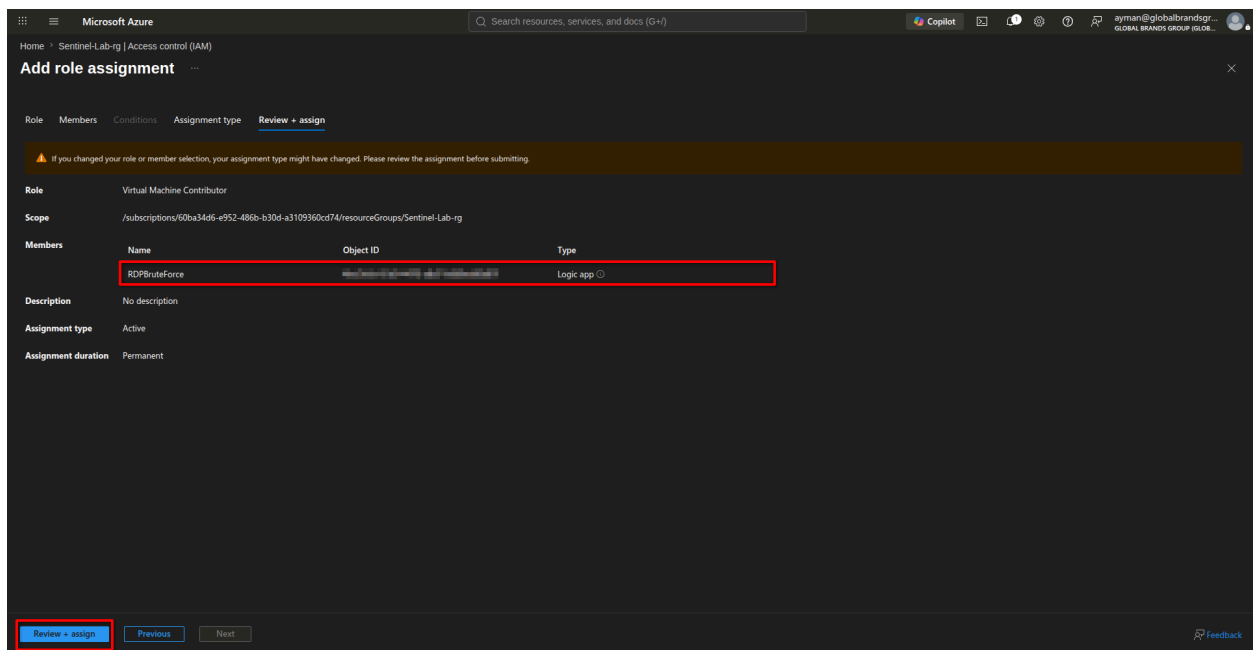
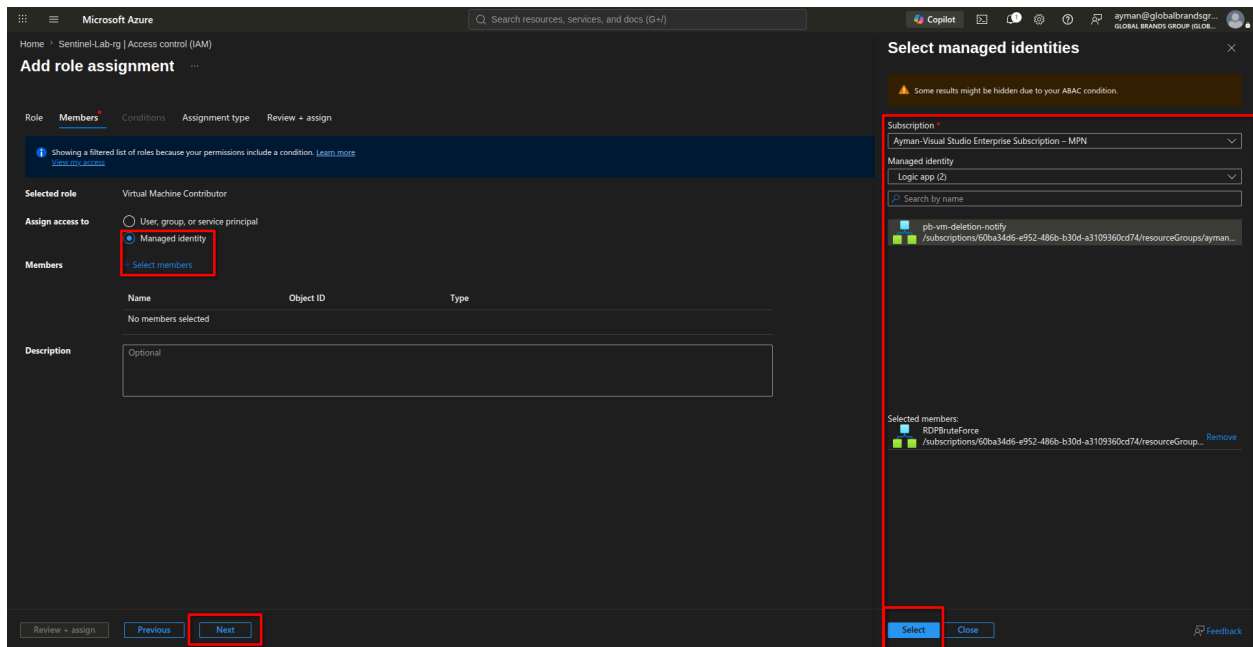
Virtual Machine Contributor

Type: All Category: All

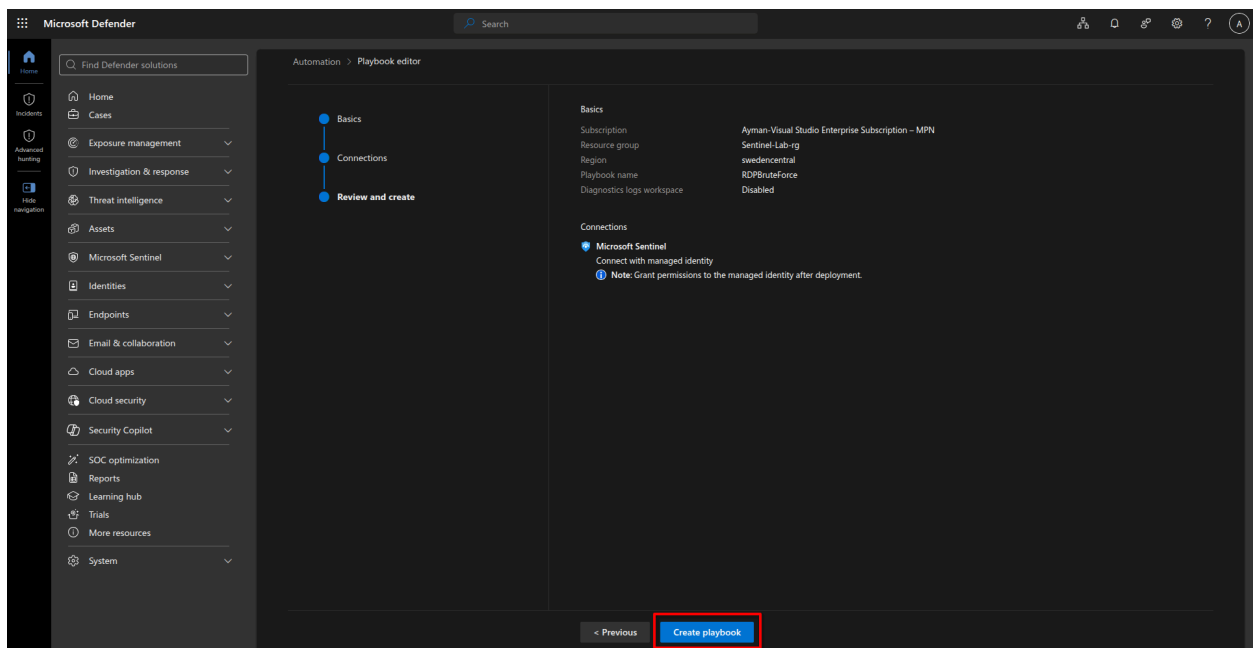
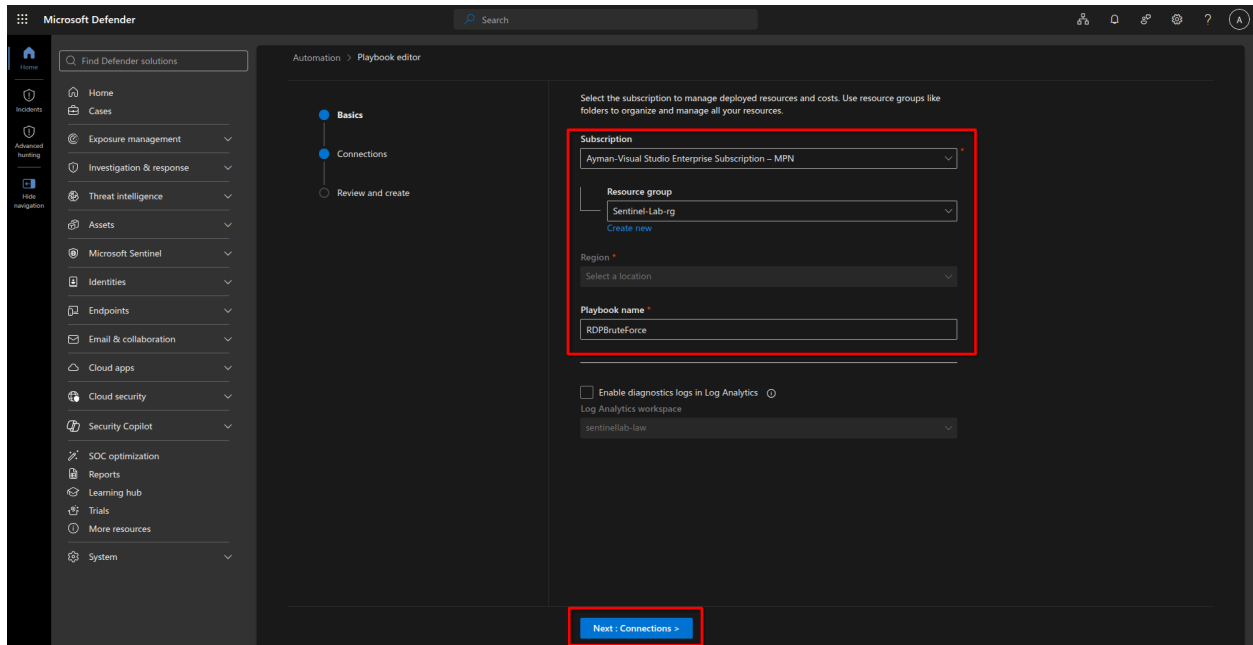
Name	Description	Type	Category	Details
Classic Virtual Machine Contributor	Lets you manage classic virtual machines, but not access to them, and not the virtual network or storage account they're connected to.	BuiltInRole	Compute	View
Desktop Virtualization Power On Contributor	Provide permission to the Azure Virtual Desktop Resource Provider to start virtual machines.	BuiltInRole	None	View
Desktop Virtualization Power On Off Contributor	Provide permission to the Azure Virtual Desktop Resource Provider to start and stop virtual machines.	BuiltInRole	None	View
Desktop Virtualization Virtual Machine Contributor	This role is in preview and subject to change. Provide permission to the Azure Virtual Desktop Resource Provider to create, delete, update, start, and stop virt...	BuiltInRole	None	View
Service Fabric Cluster Contributor	Manage your Service Fabric Cluster resources. Includes clusters, application types, application type versions, applications, and services. You will need addition...	BuiltInRole	None	View
Virtual Machine Contributor	Lets you manage virtual machines, but not access to them, and not the virtual network or storage account they're connected to.	BuiltInRole	Compute	View

Showing 1 - 6 of 6 results.

Review + assign Previous Next



3. Create Playbook

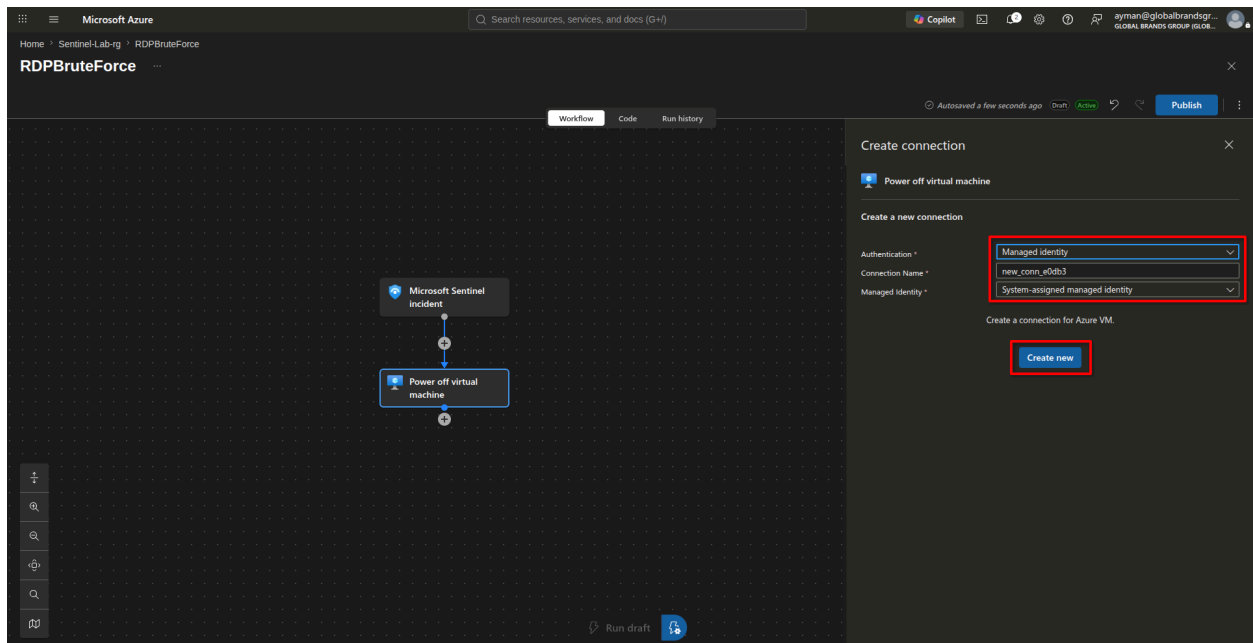
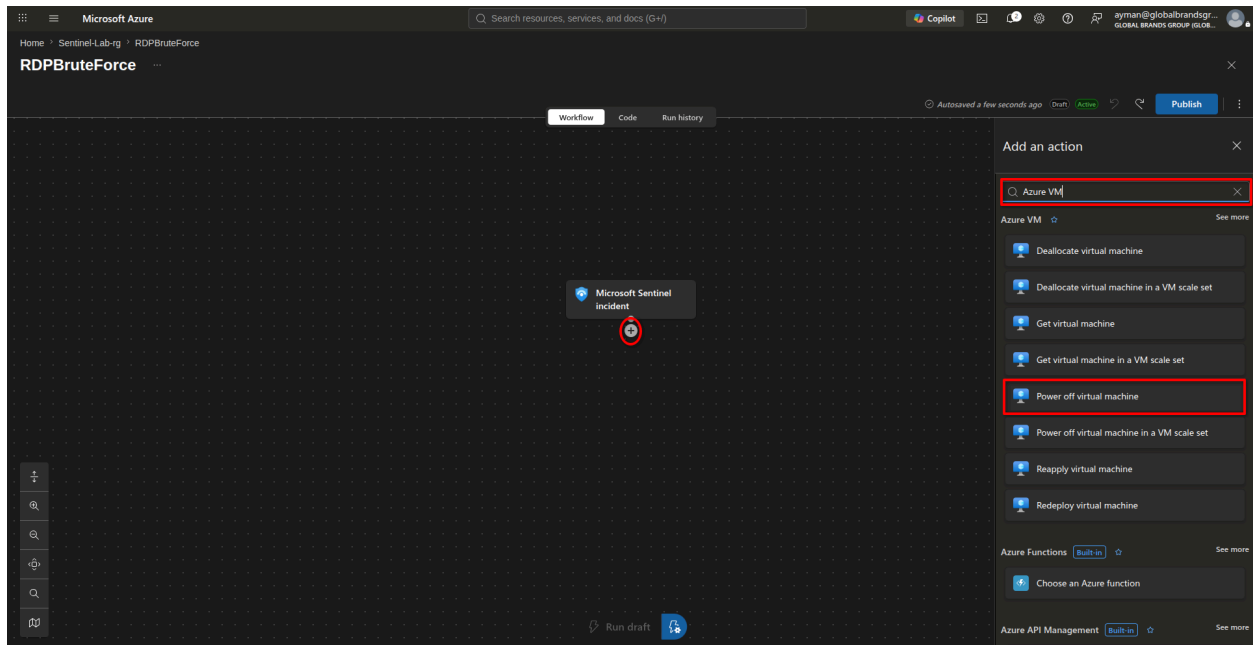


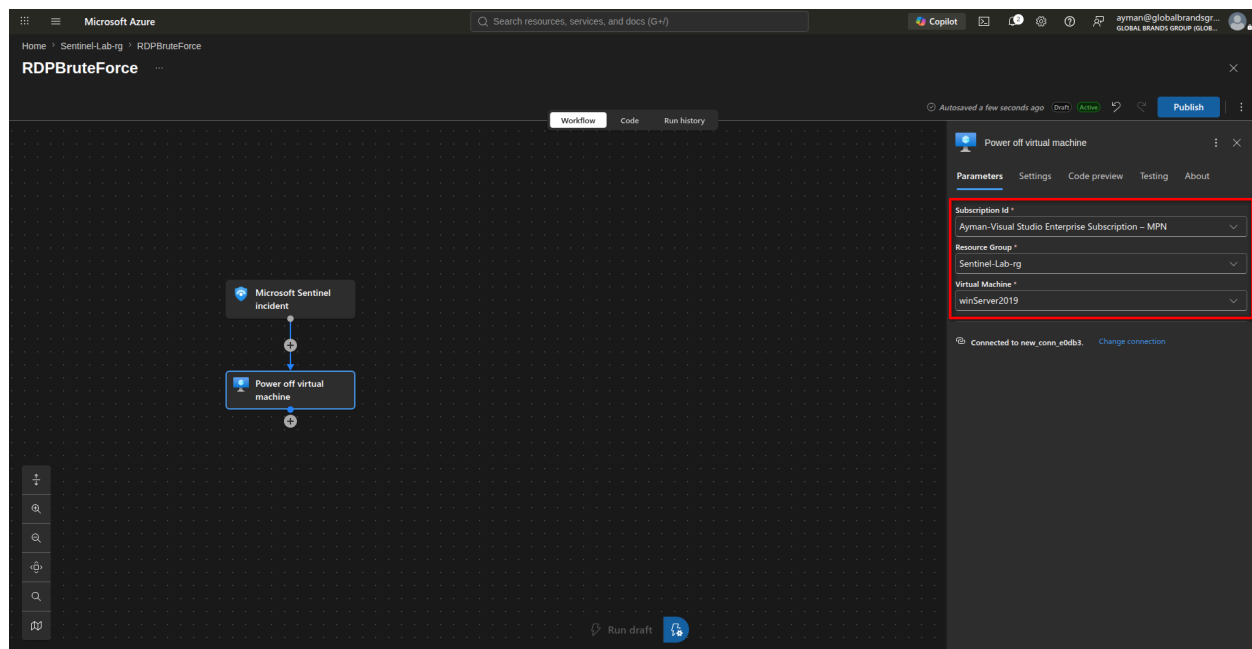
4. Use the Managed Identity in the Logic App Action

In your Logic App designer:

1. Add the **Azure VM - Power off** (or **Deallocate VM**) connector action (this is the built-in "Azure Resource Manager" connector, action: **Power off VM** or use HTTP action against Compute API)

2. When configuring the connection/authentication for that action, choose **Managed Identity** as the authentication type instead of API key or OAuth
3. Select **System-assigned** (or your user-assigned identity if you used one)





5. Get entities from the incident

After the **"Microsoft Sentinel incident"** trigger, add entity actions — these are built-in Sentinel connector actions that pull structured entity data out of the incident automatically (no manual JSON parsing needed):

1. Click the **+** below the trigger → **Add an action**
2. Search **"Microsoft Sentinel"** → look for actions like:
 - **Entities - Get Hosts**
 - **Entities - Get Accounts**
 - **Entities - Get IPs**
3. Add whichever entity types your RDP brute-force incident actually contains (likely Host + Account + IP, matching what you see in the "Microsoft Sentinel entity (Host/Account/IP)" steps you already had)

Each of these takes the trigger's **Entities** field as input (use dynamic content → **Entities** from the incident trigger) and returns an array like **Hosts**, **Accounts**, **IPs** you can reference downstream — this is what feeds your Resource Group/VM Name fields for the power-off step, and what you'll also use in the email body.

Microsoft Azure

Search resources, services, and docs (G+)

Home > Sentinel-Lab-rg > RDPBruteForce

RDPBruteForce

Workflow Code Run history

Autosaved a few seconds ago Draft Active Publish

Entities - Get Accounts

Parameters Settings Code preview Testing About

Entities list *

Entities list

Body is required.

Connected to azuresentinel-RDPBruteForce. Change connection

Microsoft Sentinel incident

Power off virtual machine

Entities - Get Accounts

Run draft

Microsoft Azure

Search resources, services, and docs (G+)

Home > Sentinel-Lab-rg > RDPBruteForce

RDPBruteForce

Workflow Code Run history

Autosaved a few minutes ago Draft Active Publish

Entities - Get Accounts

Parameters Settings Code preview Testing About

Entities list *

Entities list

Body is required.

Connected to azuresentinel-RDPBruteForce. Change connection

Microsoft Sentinel incident

Power off virtual machine

Entities - Get Accounts

Run draft

Search: enti

Microsoft Sentinel incident

Entities

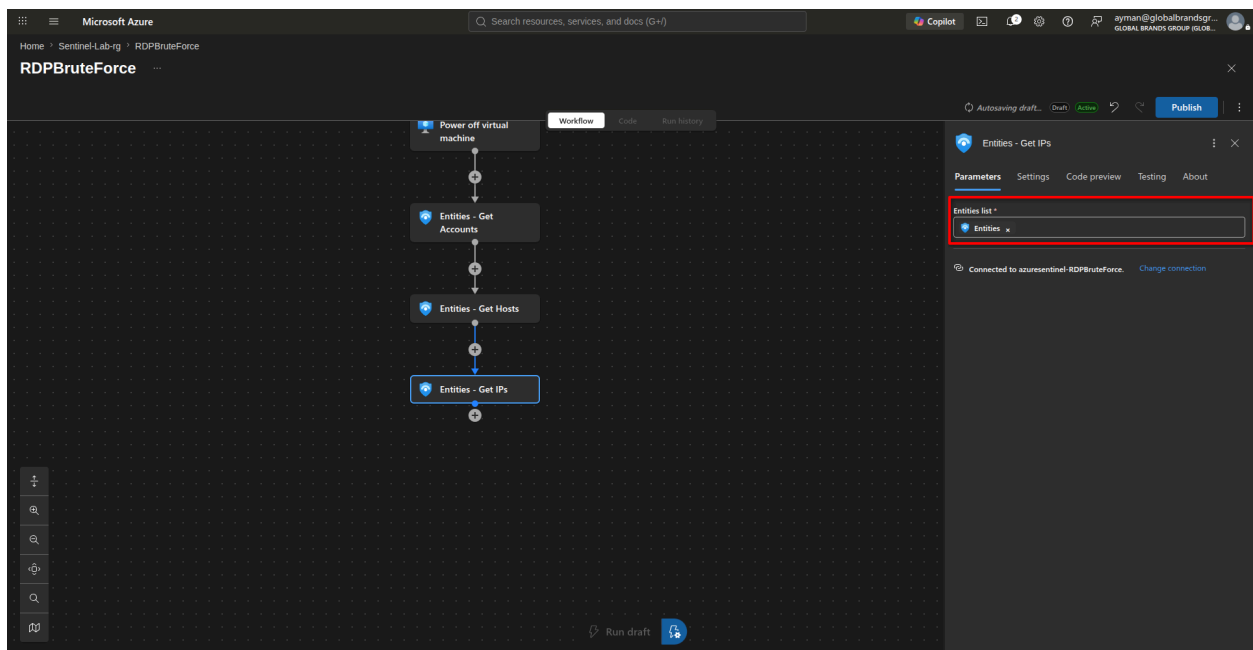
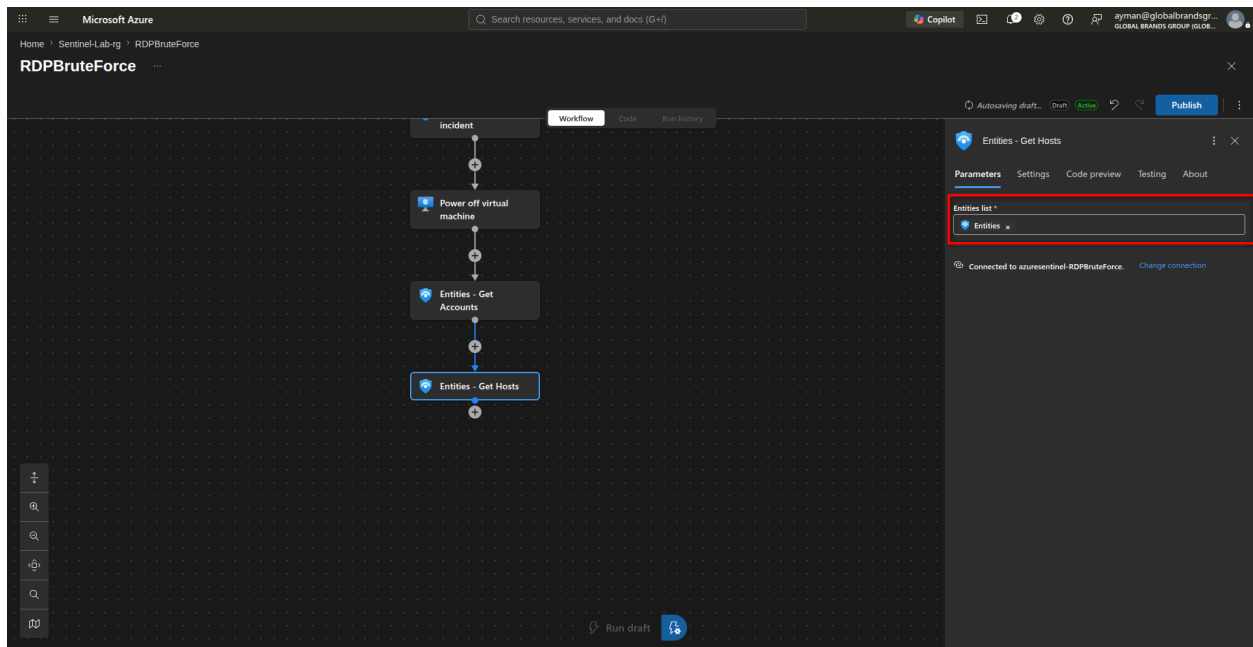
Alert Compromised Entity

Incident Sentinel ID

Incident updates Alert Compromised Entity

Incident Workspace Workspace Name

Incident Workspace Subscription ID



6. Add "Send an email" action

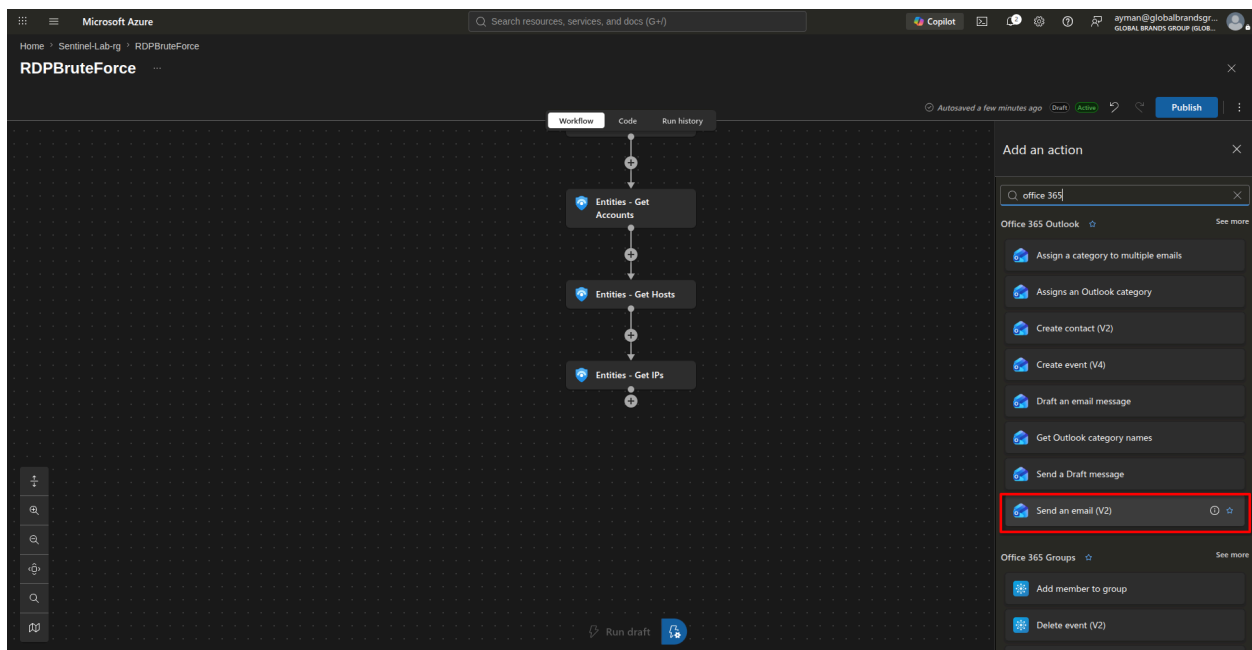
1. Click **+** → **Add an action**
2. Search **"Office 365 Outlook"** (or **Outlook.com**, or **SMTP/Gmail** depending on what mailbox you use)
3. Choose **Send an email (V2)**

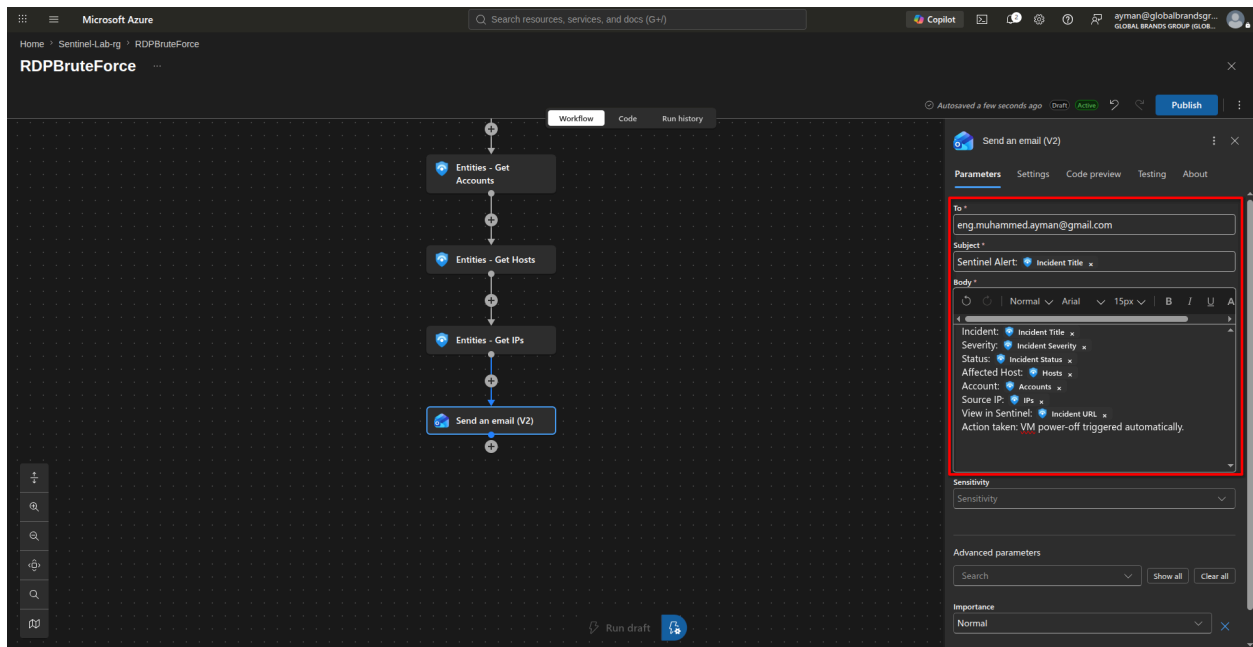
4. Configure:

- **To:** your SOC distribution list or your own address
- **Subject:** use dynamic content, e.g. `Sentinel Alert: [Incident Title]` → pull `Title` from the incident trigger
- **Body:** build it from dynamic content fields — incident Severity, Title, Incident URL (`IncidentUrl` — gives a direct link back to the Sentinel incident), plus the Host/Account/IP entity outputs from step 1 (e.g. `Hosts[0]['HostName']`, `Accounts[0]['AccountName']`, `IPs[0]['Address']`)

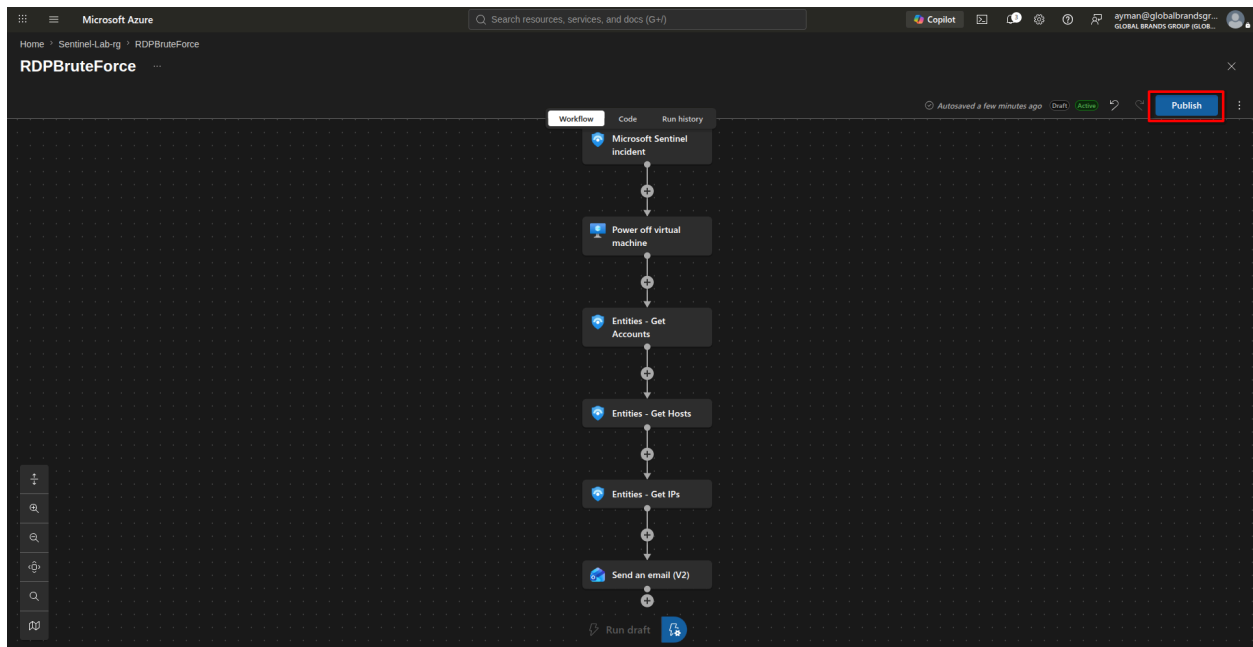
Example body (built with dynamic content, not typed literally):

```
Incident: [Title]Severity: [Severity]Status: [Status]Affected Host: [Hosts[0] HostName]Account: [Accounts[0] AccountName]Source IP: [IPs[0] Address]View in Sentinel: [IncidentUrl]Action taken: VM power-off triggered automatically.
```





7. Publish this logic app



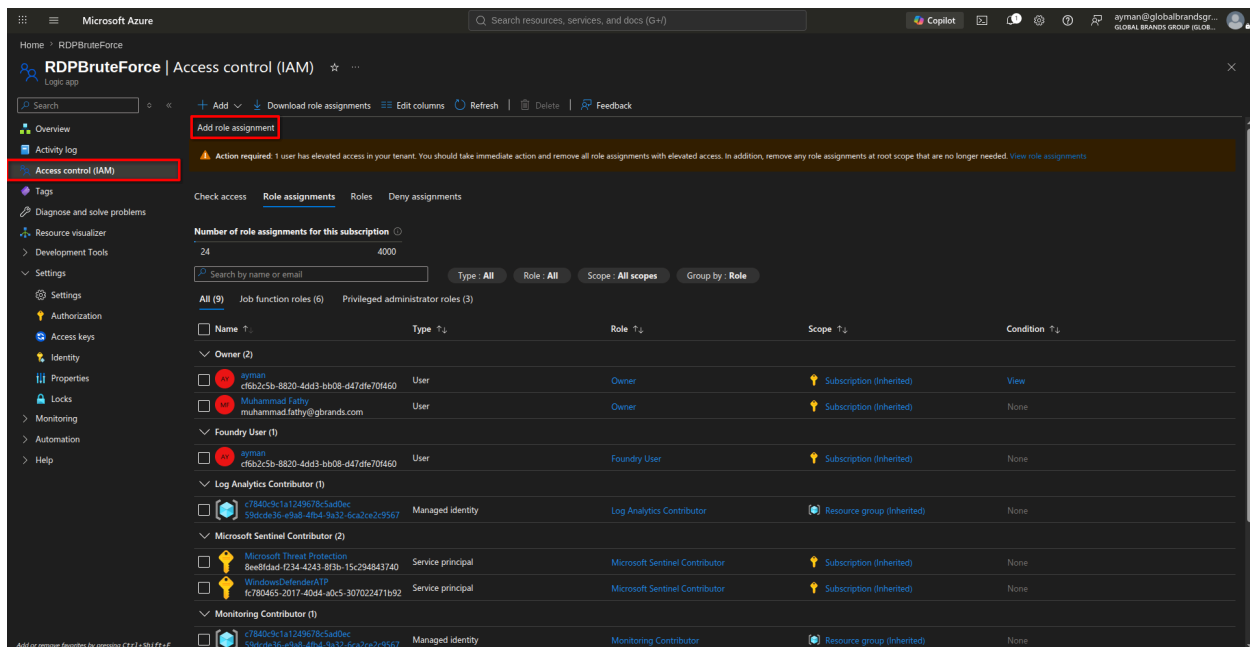
8. Grant the Logic App permission to run

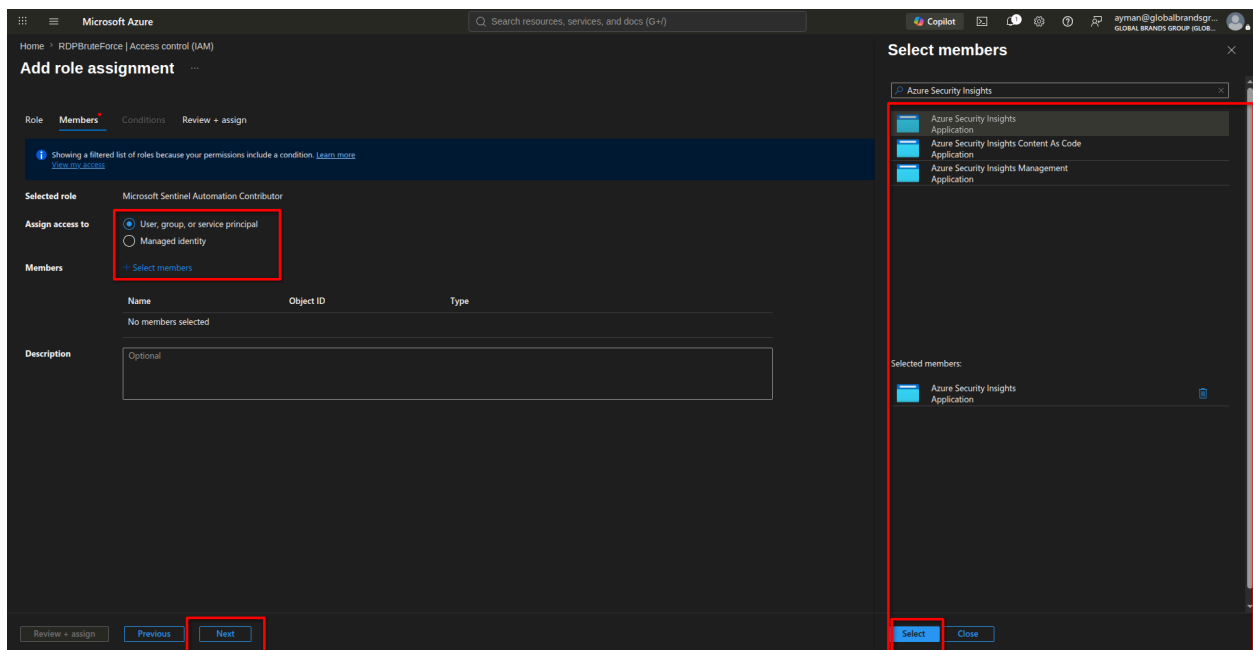
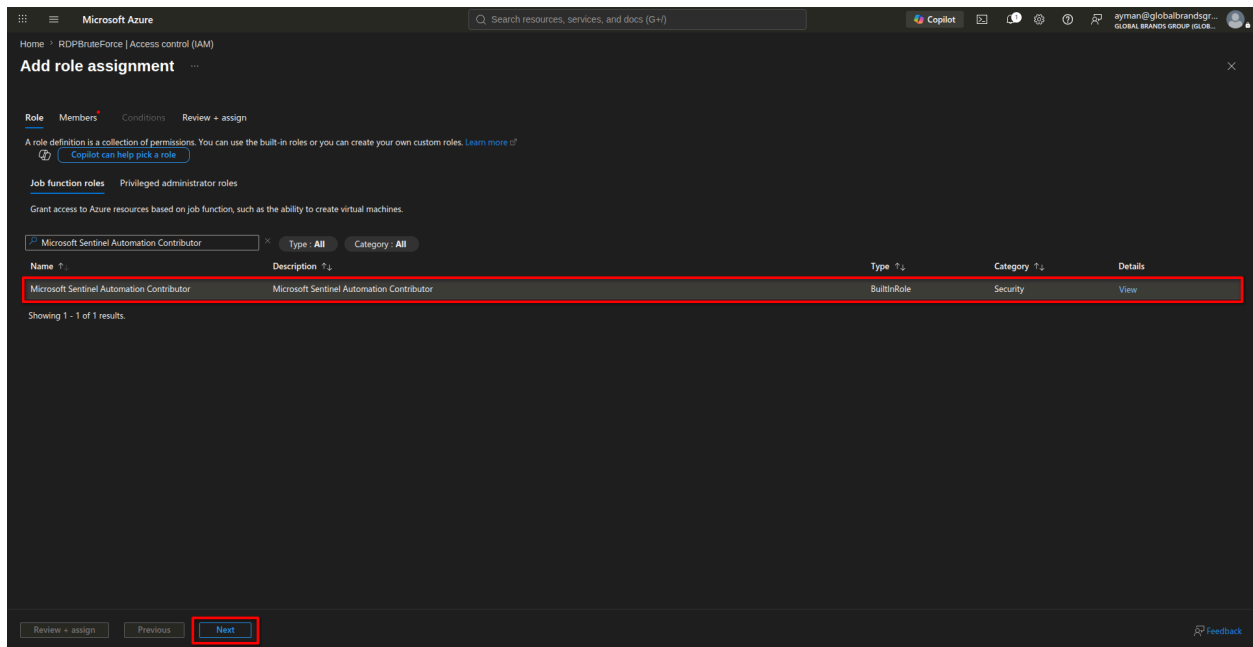
Role for Sentinel to trigger the Logic App (this is what's causing your 400 error)

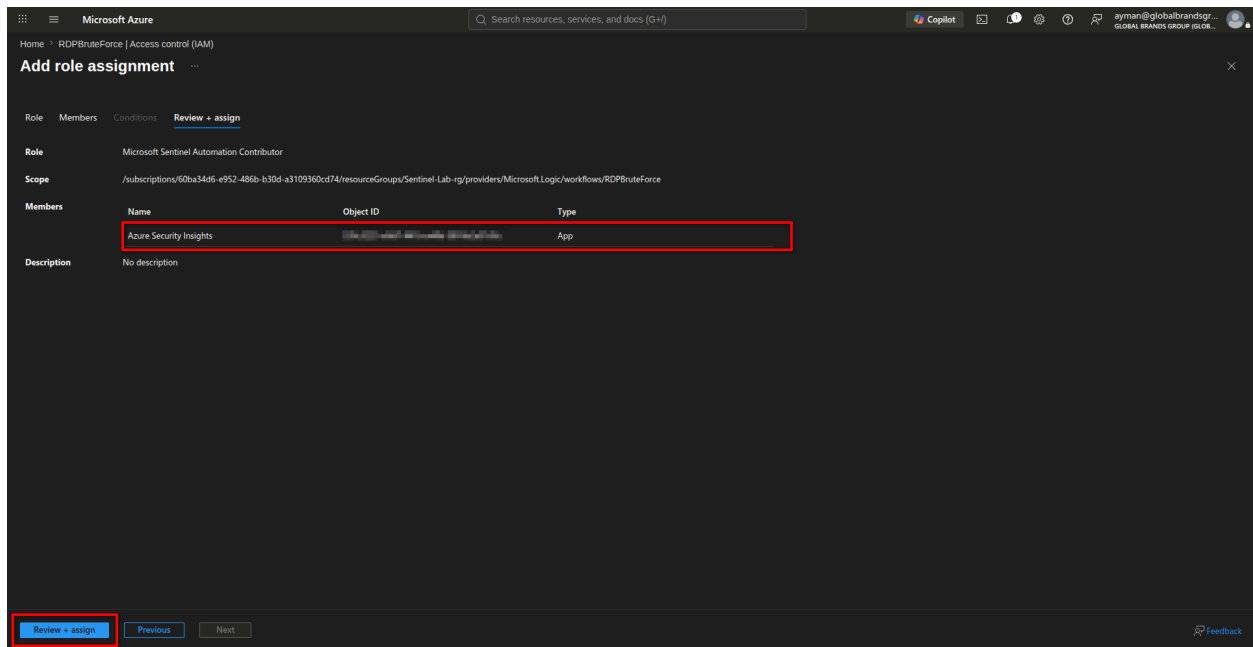
This is the opposite direction — Sentinel needs permission to *invoke* your Logic App. This is **not** assigned to the Logic App's managed identity — it's assigned to the **Microsoft Sentinel service principal** (called "Azure Security Insights" in Entra ID), scoped to the Logic App (or its Resource Group).

To fix this:

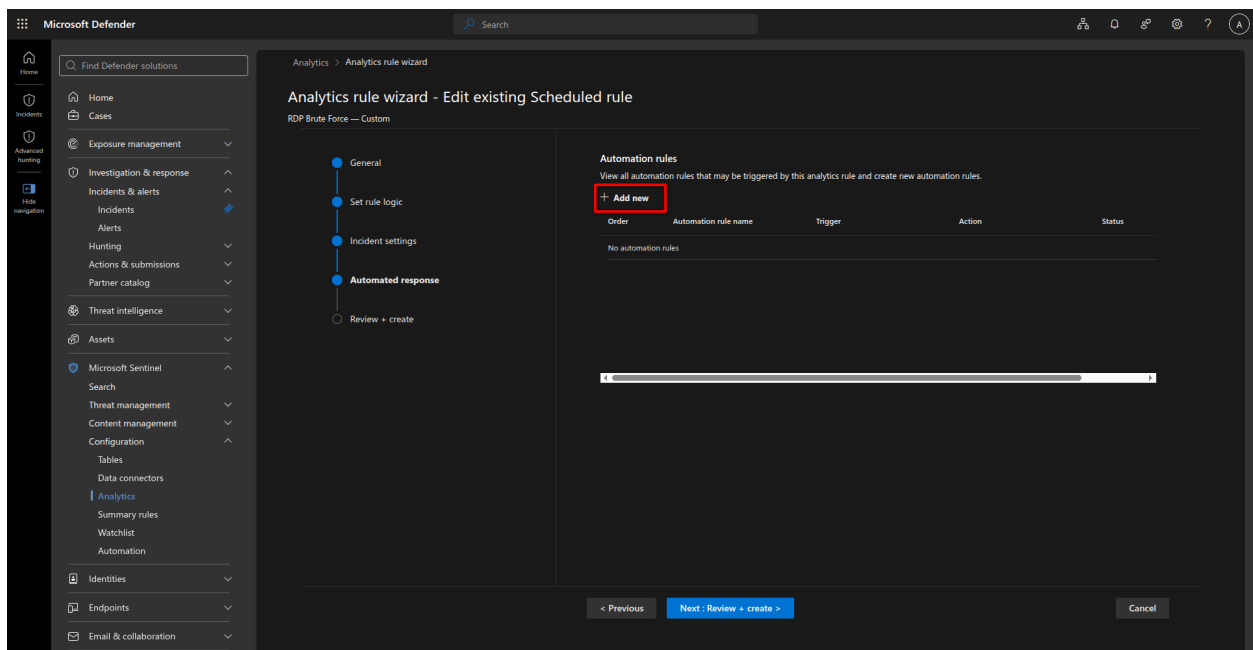
1. Go to the **Logic App (RDPBruteForce)** → **Access control (IAM)** → **Add role assignment**
2. Role: **Microsoft Sentinel Automation Contributor**
3. Assign access to: **User, group, or service principal**
4. Search for: **Azure Security Insights** (this is Sentinel's own service identity — not your user account, not the Logic App's managed identity)
5. Select it → **Review + assign**

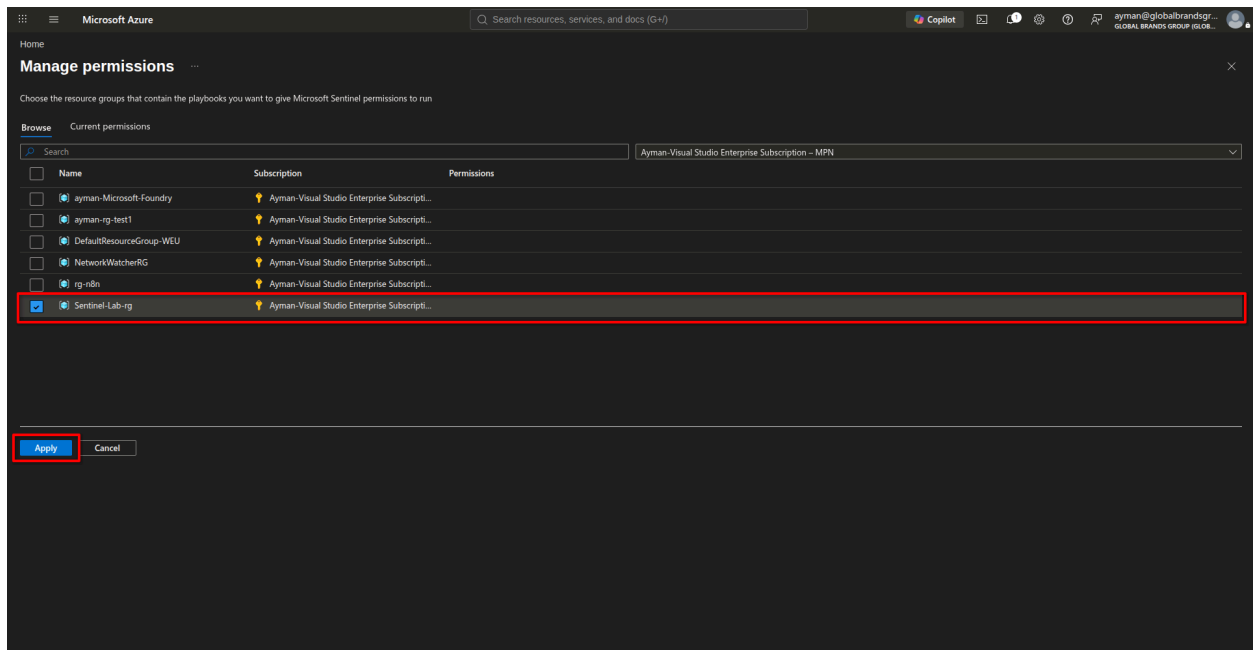
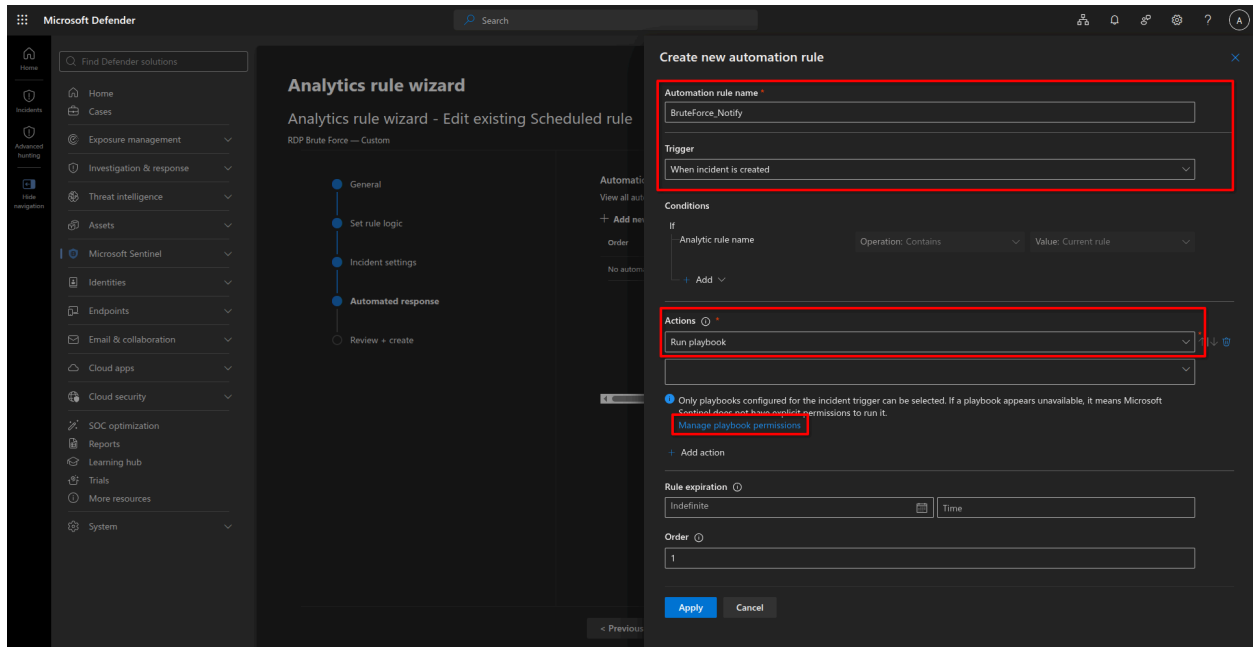


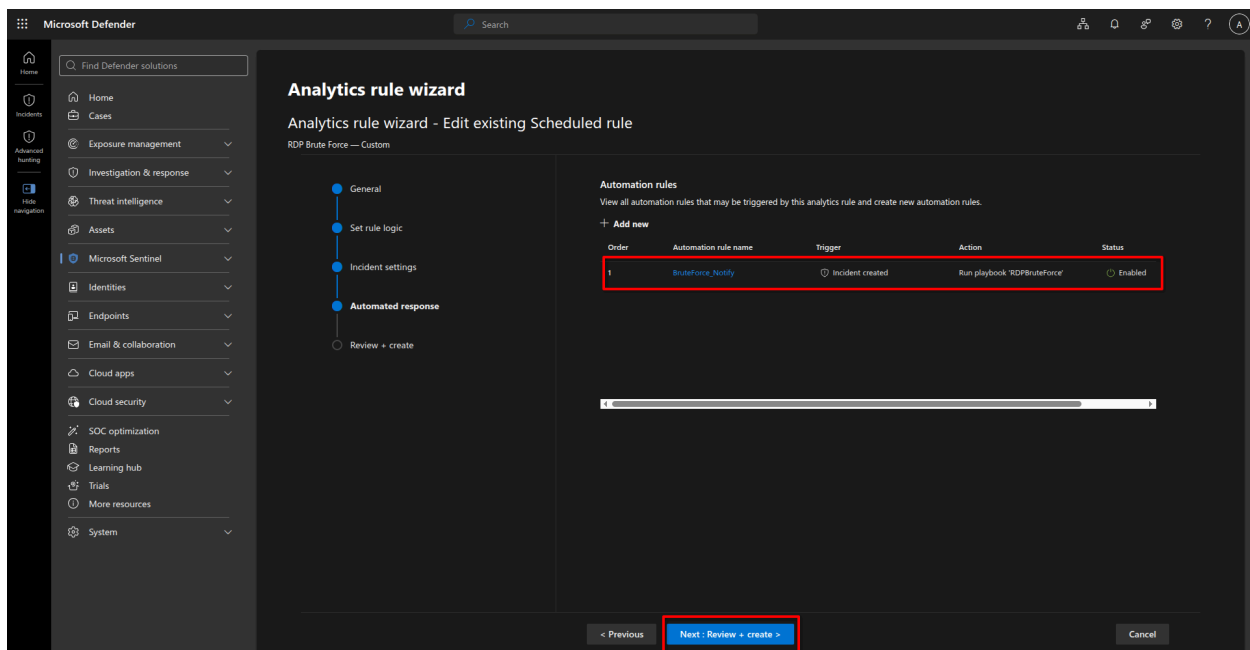
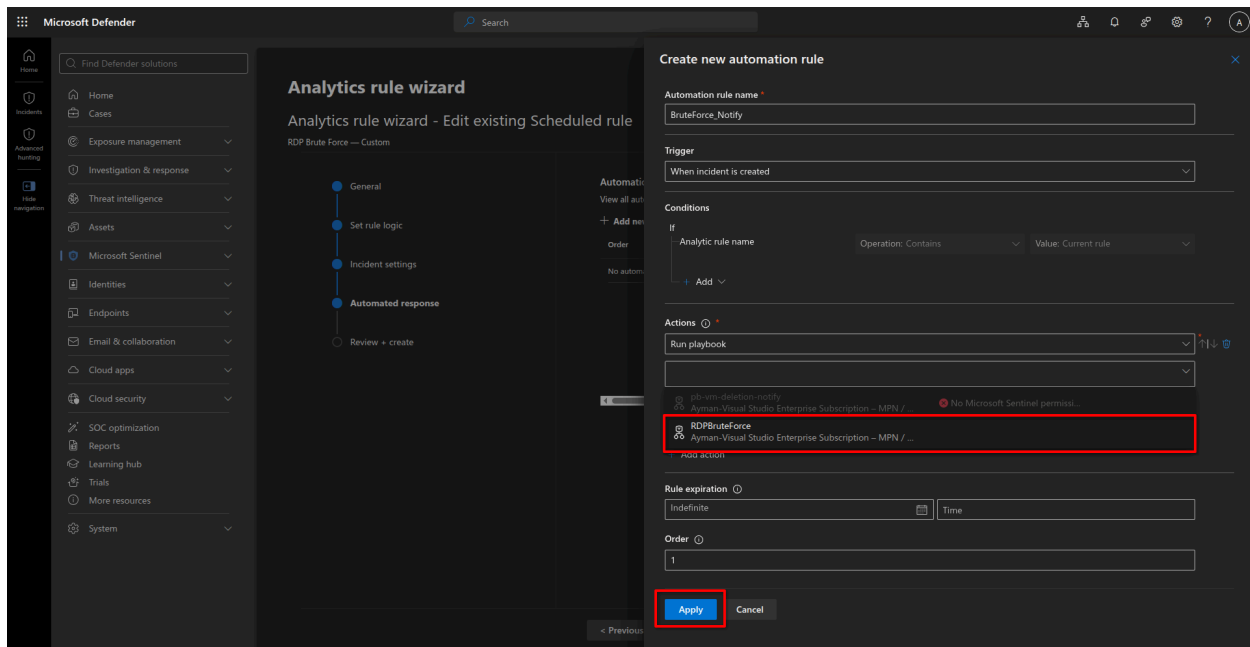




9. Edit the analytics rule to run the automation rule that triggers this playbook

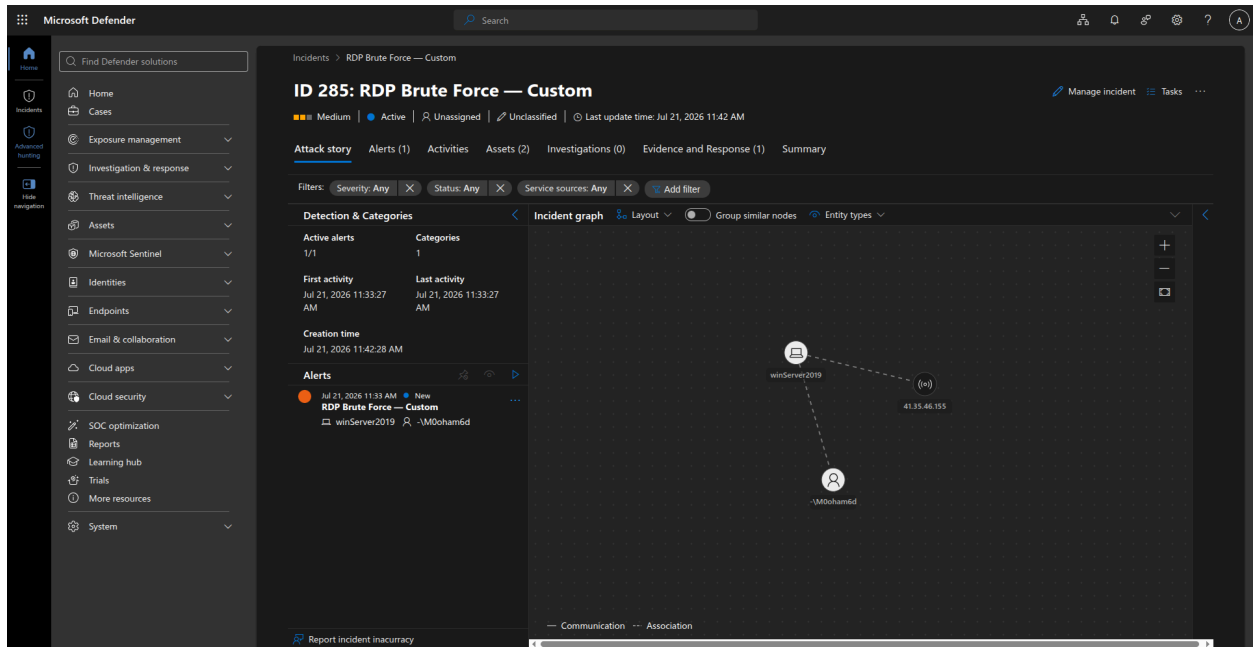




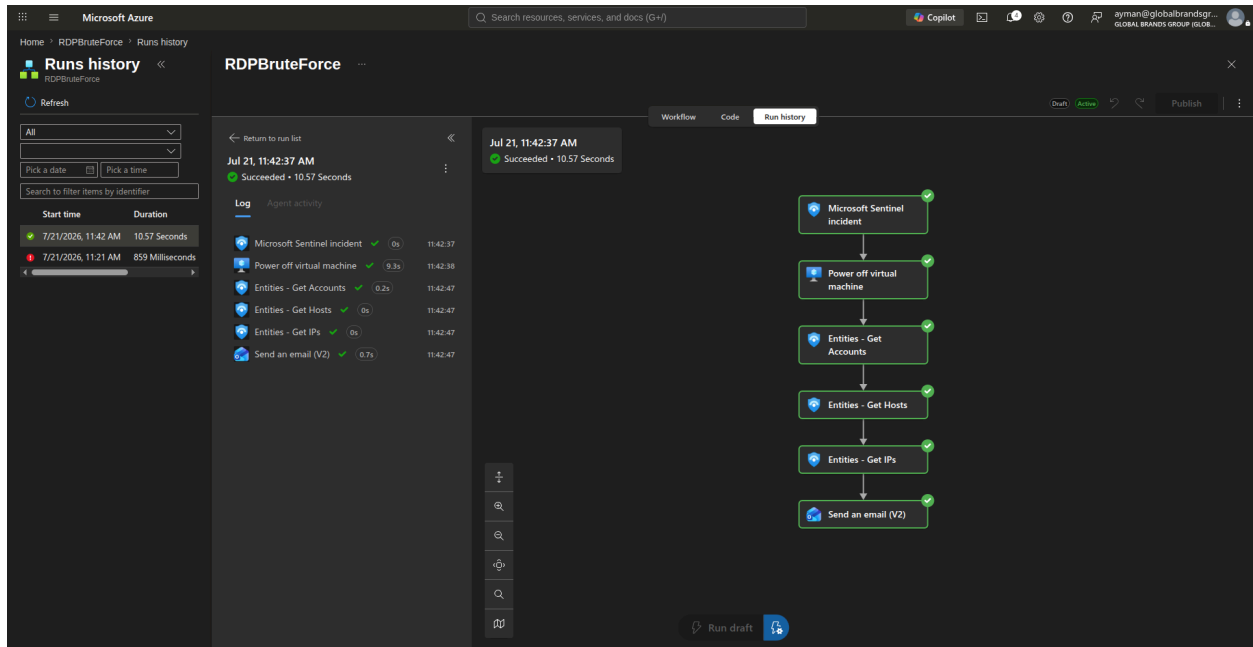


10. Testing the workflow

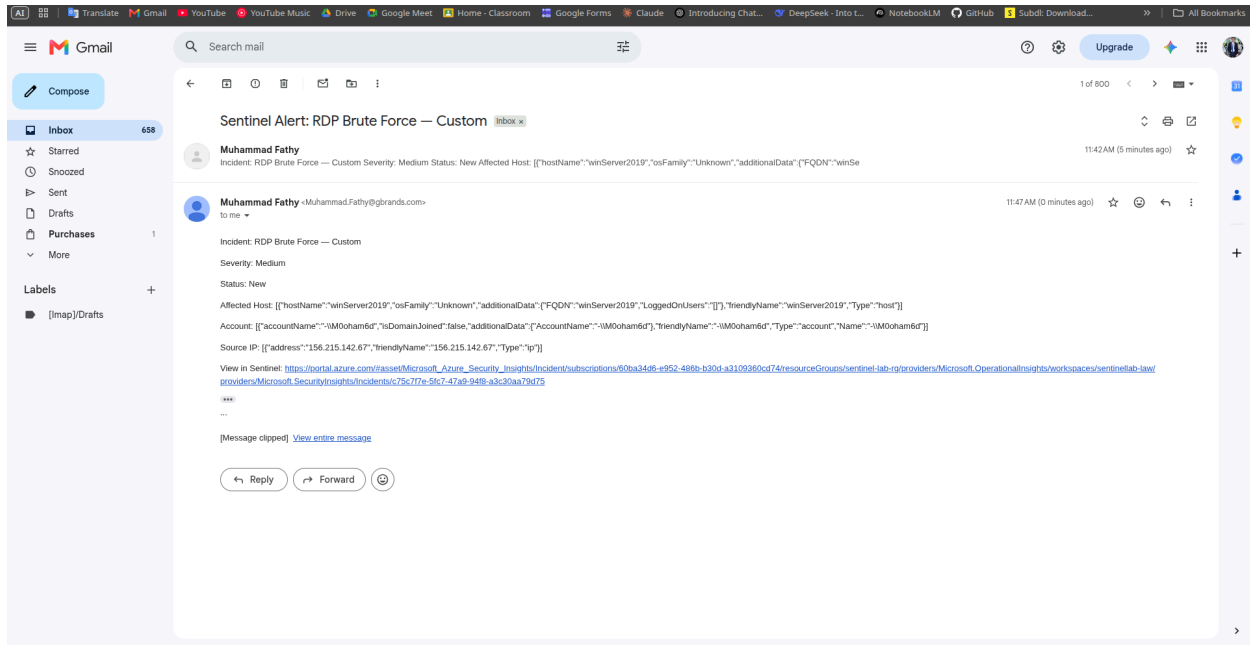
- Incident Created



- The playbook runs automatically via the automation rule.



- Email Recieved



G. Threat Intelligence & Watchlists

1. Two ways to get TI into Sentinel

Option A — TI connector (structured indicators)

- Sentinel → **Content Hub** → search "Threat Intelligence" → install the **Threat Intelligence Platforms** and/or **Threat Intelligence - TAXII** connector
- TAXII needs a feed (e.g. a free OTX or Anomaly Limo TAXII server URL + collection ID) — good if you want a live feed for the lab
- If you don't want to wire up a real feed, you can skip the connector and just add indicators manually (next step) — simpler for a lab

Microsoft Defender

484 Solutions | 365 Standalone contents | 2 Installed | 0 Updates

Search: Threat Intelligence

Status: All | Content type: All | Support: All | Provider: All | Category: All | Content sources: All

Install/Update | Delete

Content title	Status	Content source	Provider	Support
Threat Intelligence (NEW)	Not installed	Solution	Microsoft	Microsoft
Threat Intelligence	Not installed	Solution	Microsoft	Microsoft
Threat intelligence - TAXII Export	Not installed	Solution	Microsoft	Microsoft
TI Map Email entity to Cloud App Events	Not installed	Solution	Microsoft	Microsoft
TI Map Domain entity to DnsEvents	Not installed	Solution	Microsoft	Microsoft
TI Map IP entity to Web Session Events (ASIM ...	Not installed	Solution	Microsoft	Microsoft
TI Map Domain entity to PaloAlto CommonSec...	Not installed	Solution	Microsoft	Microsoft
TI Map Email entity to SecurityEvent	Not installed	Solution	Microsoft	Microsoft
TI Map Domain Entity to DeviceNetworkEvents	Not installed	Solution	Microsoft	Microsoft
TI Map IP entity to Cloud App Events	Not installed	Solution	Microsoft	Microsoft
TI Map Email entity to PaloAlto CommonSec...	Not installed	Solution	Microsoft	Microsoft

Page 1 of 1 | Showing 1 to 20 of 20 results.

Threat Intelligence (NEW)

Microsoft Provider | Microsoft Support | Version 3.0.19

Description

Note: Please refer to the following before installing the solution:

- Review the solution Release Notes
- There may be known issues pertaining to this Solution, please refer to them before installing.

Microsoft Sentinel has recently improved its threat intelligence hunting experience by incorporating support for STIX objects like Threat Actor, Attack Pattern, Identity, and Relationship. As a result, we have updated our TI Solutions to leverage the new ThreatIntelligence table. Work with STIX objects and indicators to enhance threat intelligence and threat hunting in Microsoft Sentinel (Preview) - Microsoft Sentinel | Microsoft Learn.

The Threat Intelligence solution contains data connectors for import of supported STIX objects into Microsoft Sentinel, analytic rules for matching TI data with event data, workbook, and hunting queries. Threat indicators can be malicious IP's, URL's, filehashes, domains, email addresses etc.

Data Connectors: 6, Parsers: 1, Workbooks: 1, Analytic Rules: 52, Hunting Queries: 5

Content type: 53 Analytics rule, 6 Data connector, 5 Hunting query

1 Parser, 1 Workbook

Category: Security - Threat Intelligence

Pricing: Free

Install | View details

Microsoft Defender

Threat Intelligence (NEW) > Data connectors

Refresh | Guides | Workspace: sentinellab-lab

Device specific AMA connectors have been deprecated. Learn more

In anticipation of the End of Support of the HTTP Data Collector API, Azure Function-based data connectors using legacy API will not be supported after September 14, 2026. Please update to GCP-based data connectors. Learn more

10 Connectors | 4 Connected | More content at Content Hub

Search by name or provider | Providers: Microsoft

Data Types: All | Status: All

Status	Connector name	Updates
Connected	Azure Activity	...
Connected	Microsoft 365 Insider ...	...
Connected	Microsoft Defender T...	...
Connected	Premium Microsoft D...	...
Disconnected	[DEPRECATED] Secur...	Deprecated
Disconnected	Threat intelligence - T...	...
Disconnected	Threat intelligence - T...	...
Disconnected	Threat Intelligence Pla...	...
Disconnected	Threat Intelligence Up...	...
Disconnected	Windows Security Eve...	...

Threat Intelligence Platforms

Disconnected Status | Microsoft Provider | Last Log Received

Description

Microsoft Sentinel integrates with Microsoft Graph Security API data sources to enable monitoring, alerting, and hunting using your threat intelligence. Use this connector to send threat indicators to Microsoft Sentinel from your Threat Intelligence Platform (TIP), such as Threat Connect, Palo Alto Networks MindMeld, MISP or other integrated applications. Threat indicators can include IP addresses, domains, URLs, and file hashes. For more information, see the Microsoft Sentinel documentation >.

Last data received: --

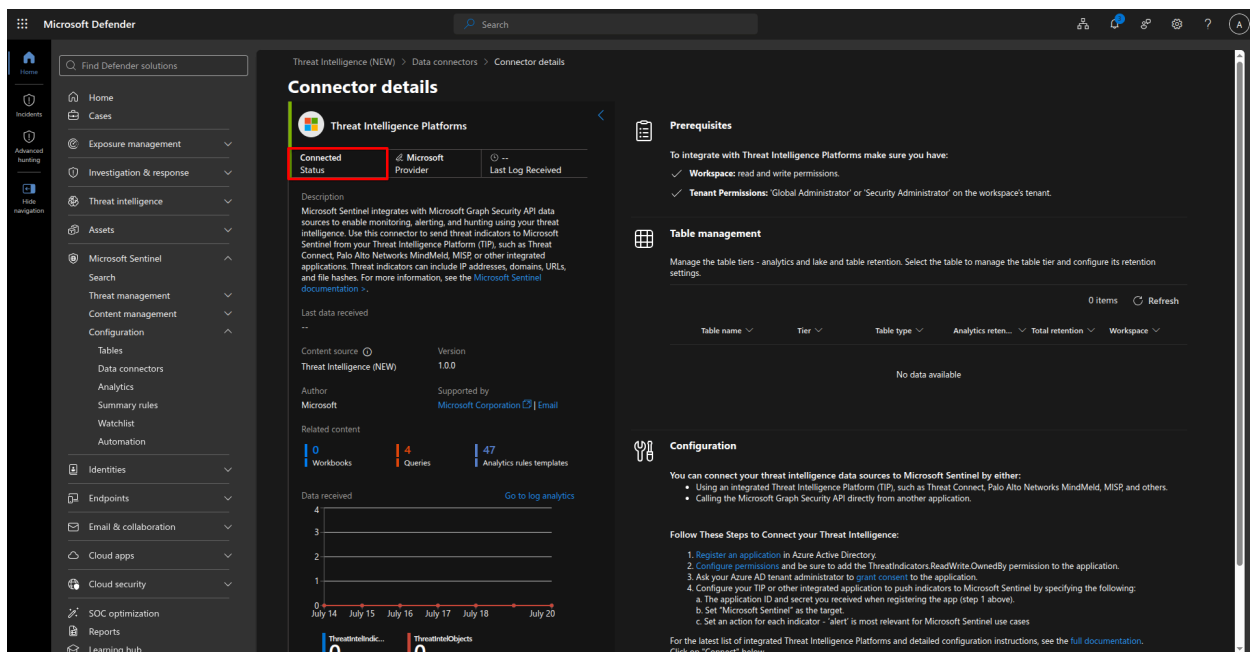
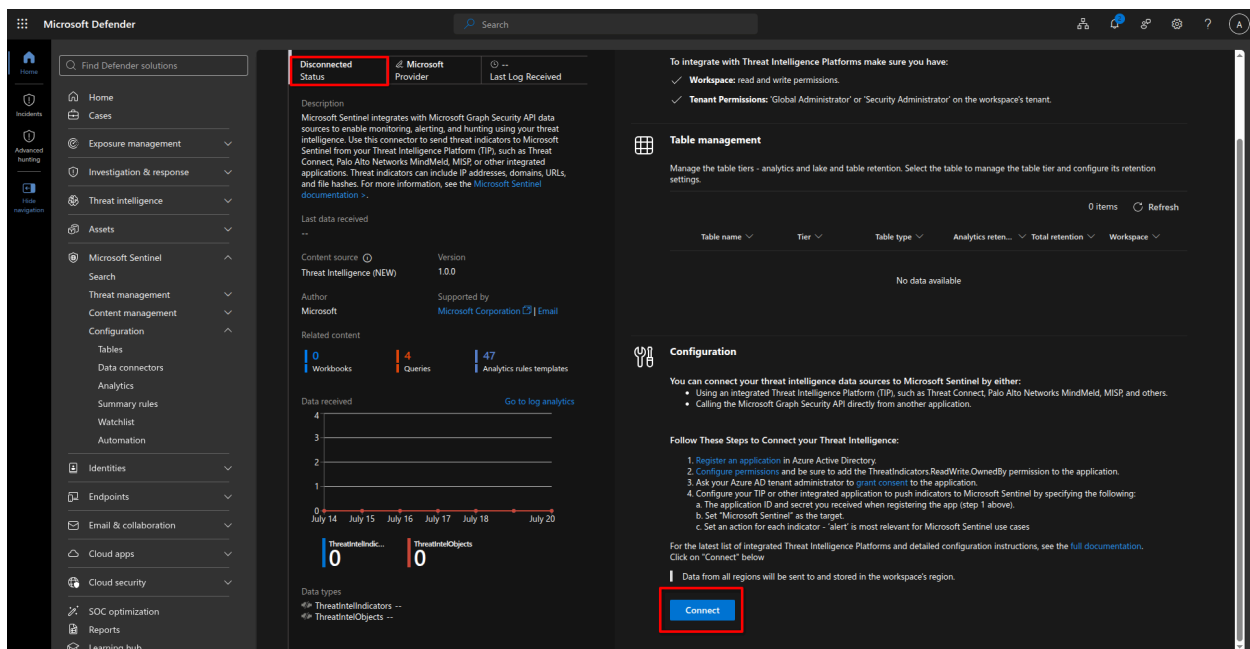
Content source: Threat Intelligence (NEW) | Version: 1.0.0

Author: Microsoft | Supported by: Microsoft Corporation | Email

Related content: 0 Workbooks, 4 Queries, 47 Analytics rules templates

Data received: 4 | Go to log analytics

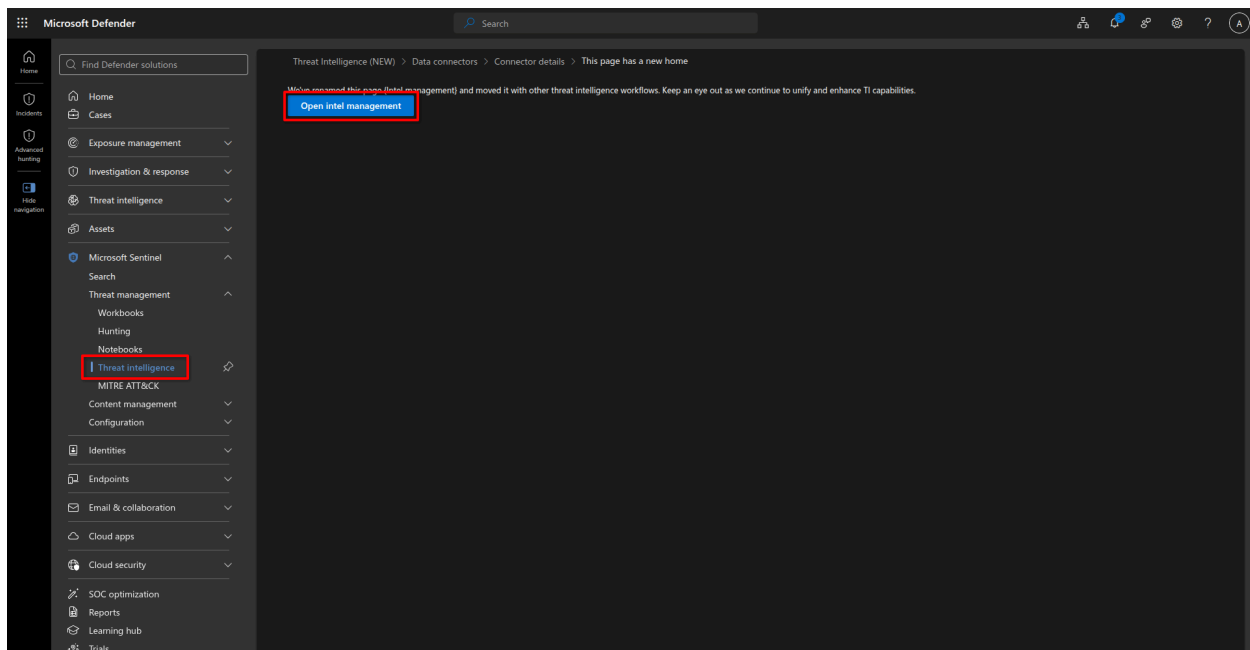
Open connector page

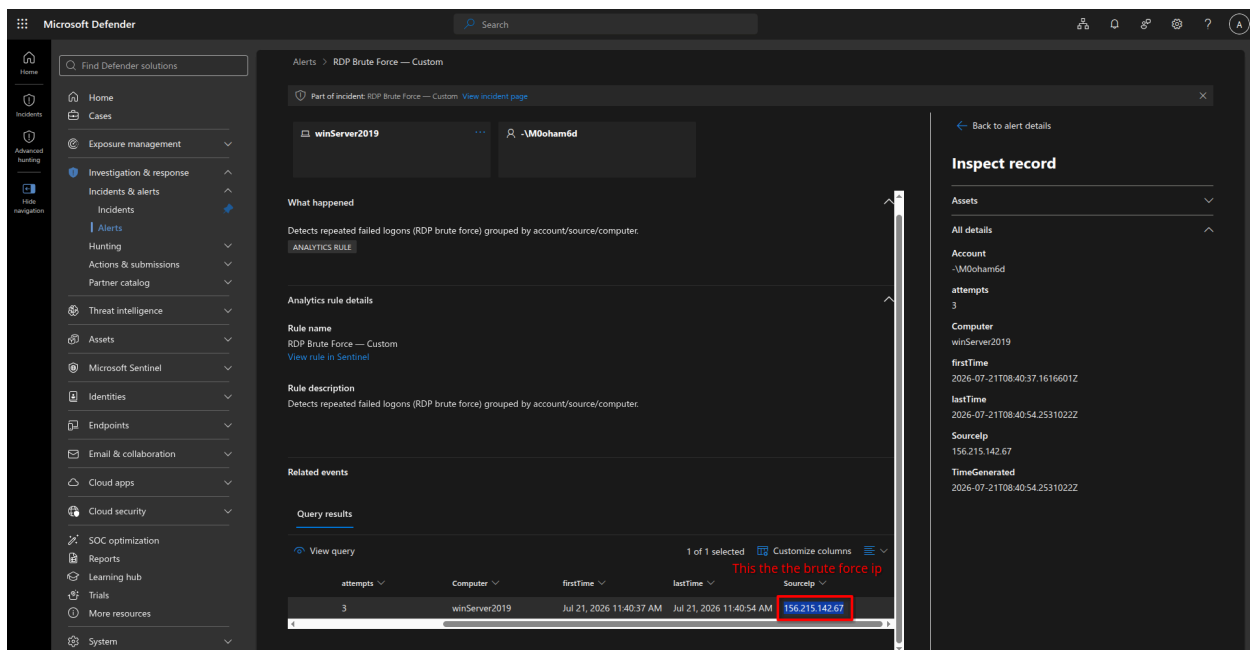
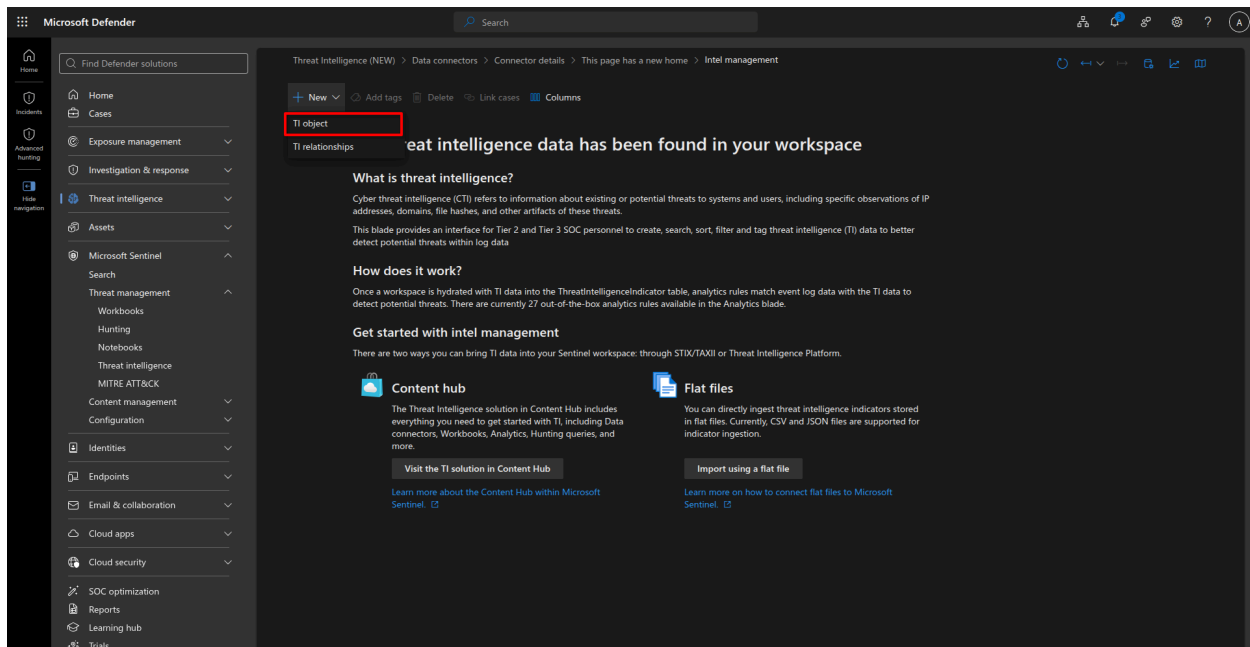


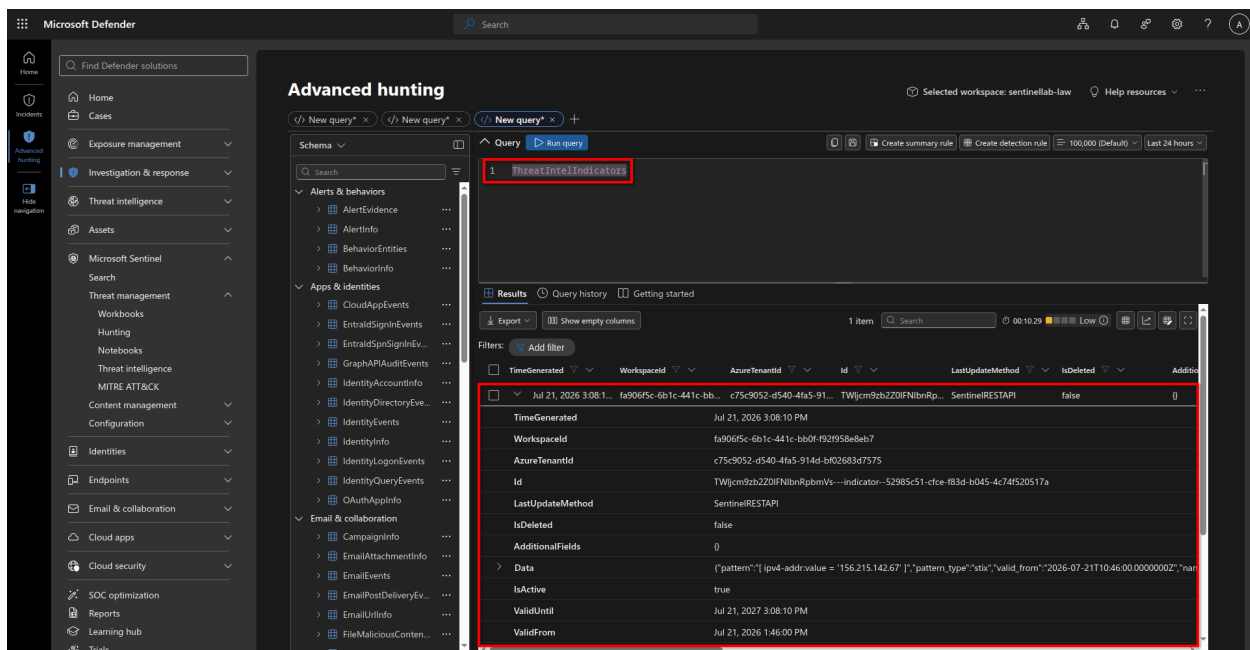
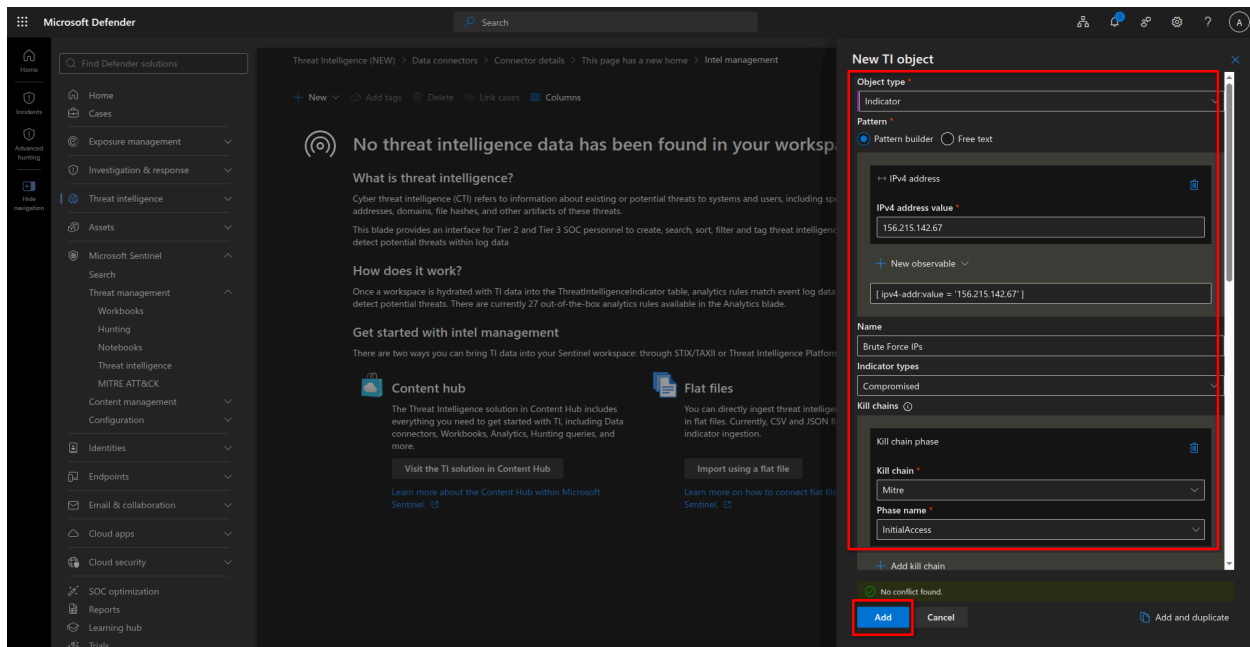
Option B — Manual indicators (fastest for testing)

- Sentinel → **Threat Intelligence** (left menu, under Threat Management)
- **+ Add new** → indicator type (IP, File hash, Domain, URL)
- Add a test indicator: e.g. IP **20.20.20.20**, confidence, threat type = "brute-force source", expiration date

- These land in the `ThreatIntelIndicators` table
 - Microsoft retired official support for the legacy `ThreatIntelligenceIndicator` table on 31 May 2026, replacing it with two new STIX-based tables. Since your indicator was created **after** that retirement, it never lands in the old table — that's why `ThreatIntelligenceIndicator` returns 0 results.
 - The new tables are `ThreatIntelIndicators` (IoCs, structured like the old table but richer) and `ThreatIntelObjects` (Threat Actor, Attack Pattern, Relationship, Identity objects). The `ThreatIntelIndicators` table is now what stores all your Sentinel threat indicators.







2. Create a Watchlist (simpler, CSV-based)

1. Prep a CSV file

Create a small CSV locally, e.g. `knownbadips.csv`:

```
IPAddress,Reason,AddedBy,DateAdded
156.215.142.67,Brute force source - lab TI indicator,Muhammad,2026-07-21
203.0.113.45,Repeated failed RDP attempts,Muhammad,2026-07-21
198.51.100.22,Known scanner - Shodan-like behavior,Muhammad,2026-07-21
```

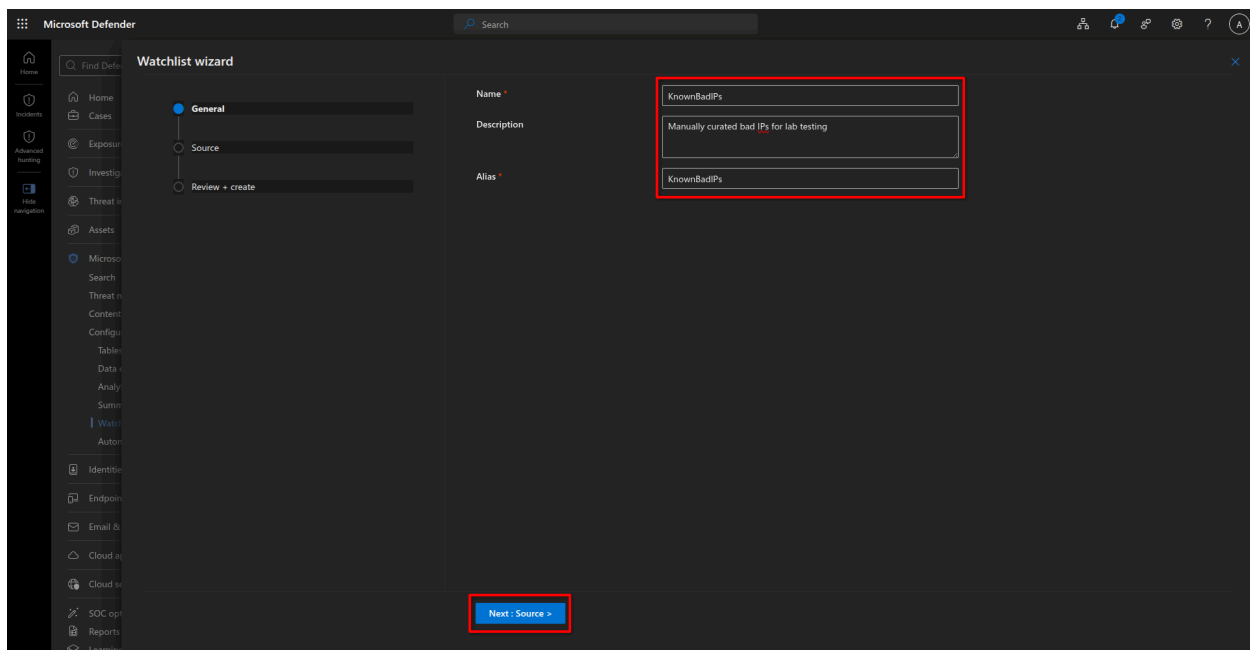
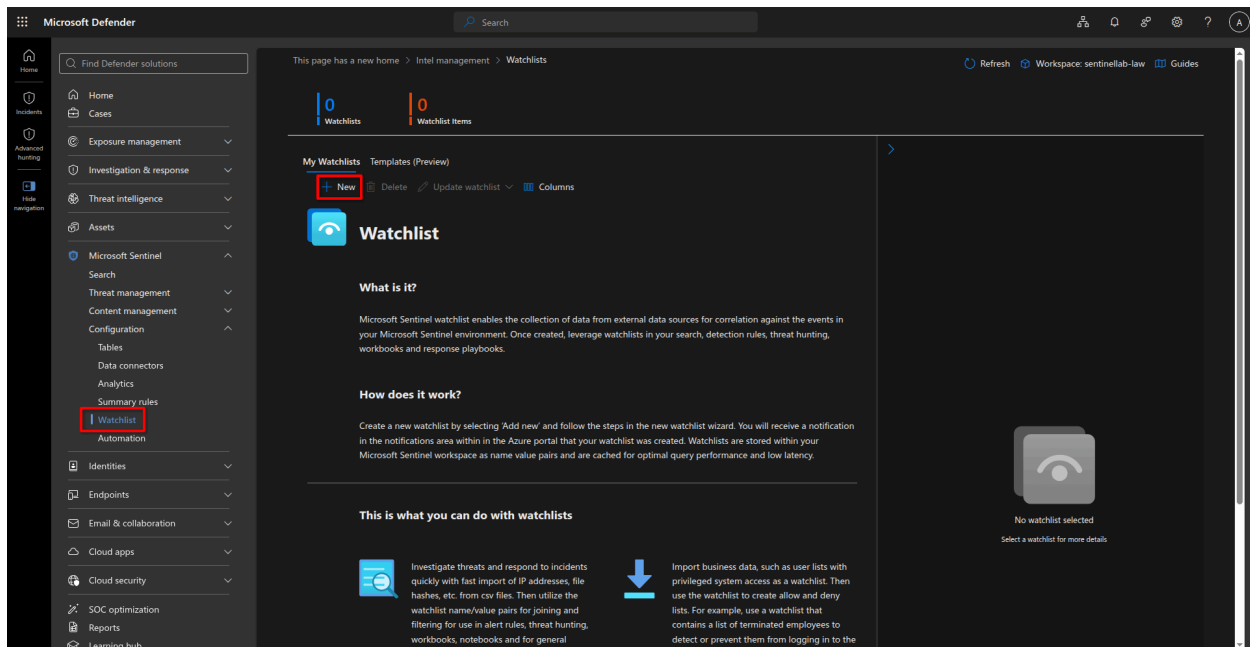
Header row required — first row becomes your column names.

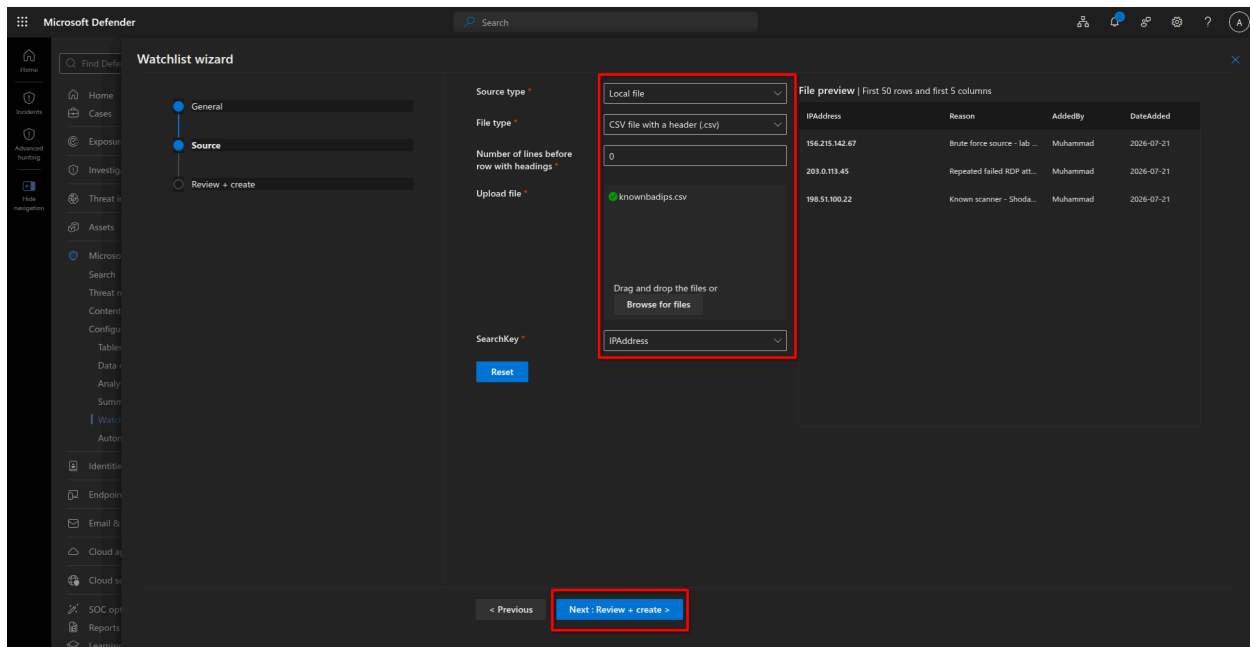
2. Create it in the portal

Defender portal (unified, since that's what you're in):

1. Left nav → **Microsoft Sentinel** → **Configuration** (or under Threat management, depending on your layout) → **Watchlists**
2. **+ Add new**
3. Fill in the wizard:
 - **Name:** `KnownBadIPs`
 - **Alias:** `KnownBadIPs` (this is what you reference in KQL)
 - **Description:** "Manually curated bad IPs for lab testing"
4. **Source:** choose **Upload file (CSV)** → browse to your CSV
5. Set **SearchKey** = `IPAddress` (this is the column used for fast lookups)
6. Review + Create — it takes a minute or two to provision

Watchlists are best for stuff you curate yourself (VIP accounts, known-bad IPs, allow-listed service accounts) — they query much faster than joining the full TI table.

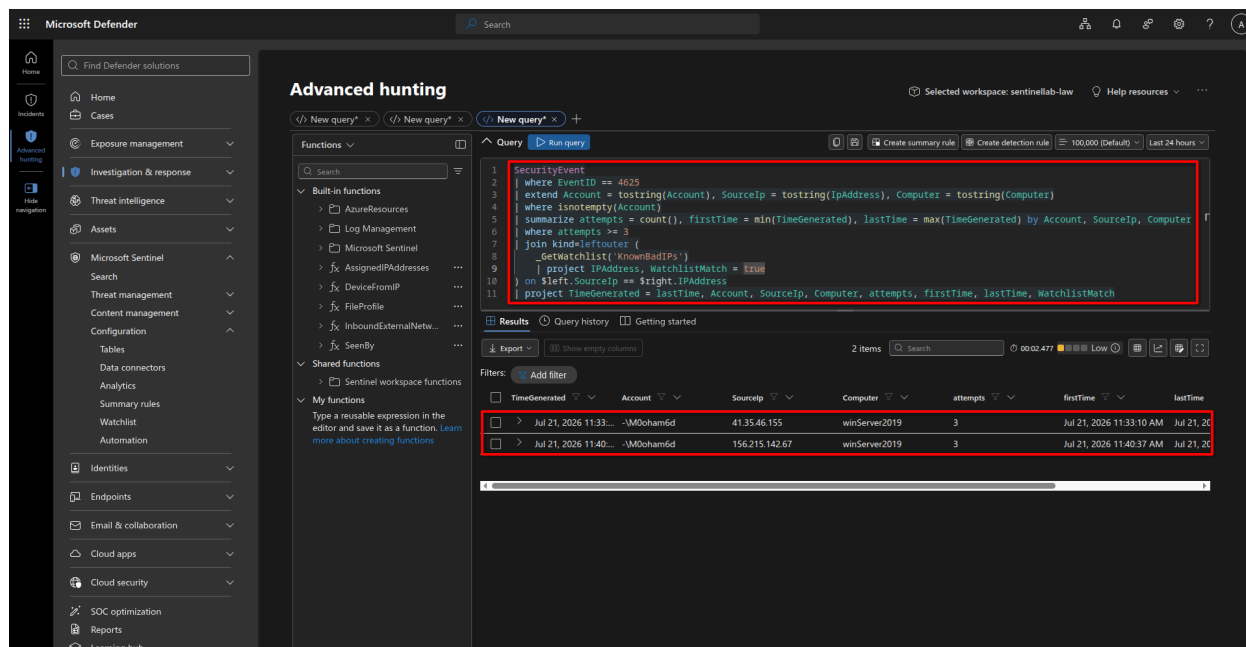




3. Use TI/watchlist in your RDP brute-force rule

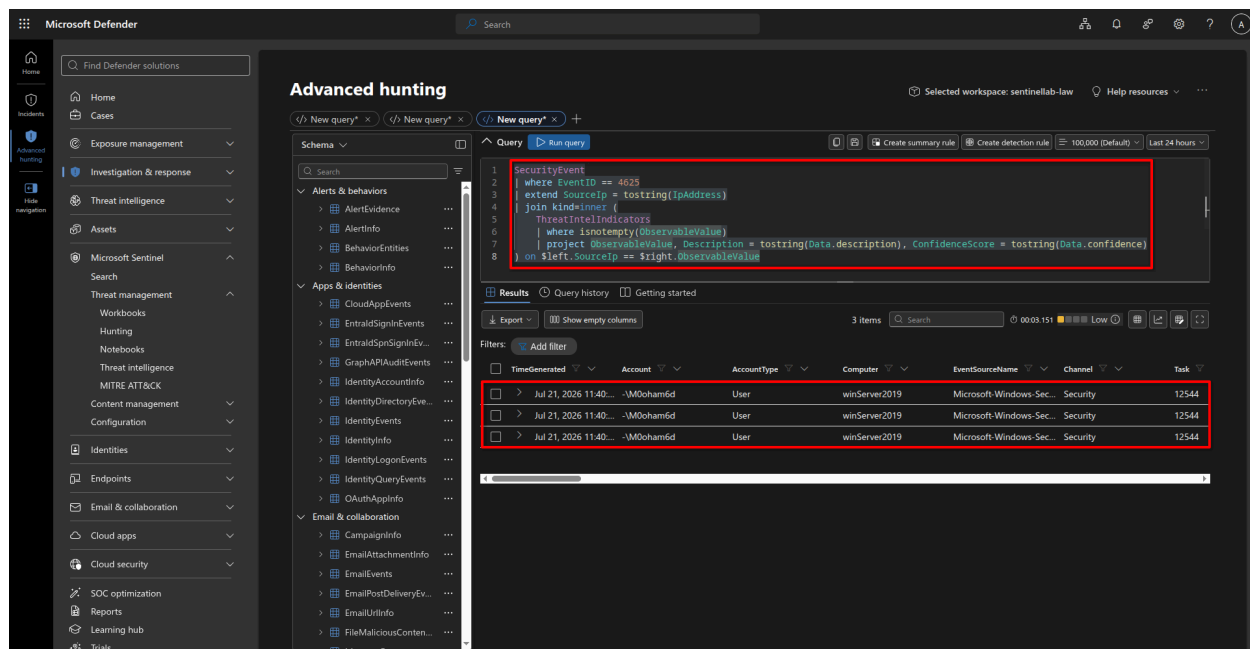
Update the KQL from section D to enrich matches against your watchlist:

```
SecurityEvent
| where EventID == 4625
| extend Account = tostring(Account), SourceIp = tostring(IpAddress), Computer = tostring(Computer)
| where isnotempty(Account)
| summarize attempts = count(), firstTime = min(TimeGenerated), lastTime = max(TimeGenerated) by Account, SourceIp, Computer
| where attempts >= 3
| join kind=leftouter (
    _GetWatchlist('KnownBadIPs')
    | project IPAddress, WatchlistMatch = true
) on $left.SourceIp == $right.IPAddress
| project TimeGenerated = lastTime, Account, SourceIp, Computer, attempts, firstTime, lastTime, WatchlistMatch
```



Or join against the actual TI table instead of a watchlist:

```
SecurityEvent
| where EventID == 4625
| extend SourceIp = tostring(IPAddress)
| join kind=inner (
    ThreatIntelIndicators
    | where isnotempty(ObservableValue)
    | project ObservableValue, Description = tostring(Data.description), ConfidenceScore = tostring(Data.confidence)
) on $left.SourceIp == $right.ObservableValue
```



H. Notebooks (Jupyter + MSTICPy)

1. Access the Notebooks blade

Azure Portal:

1. Microsoft Sentinel → left menu → **Threat Management** → **Notebooks**
2. You'll see two tabs: **Templates** (Gallery) and **My notebooks**

Defender portal (unified):

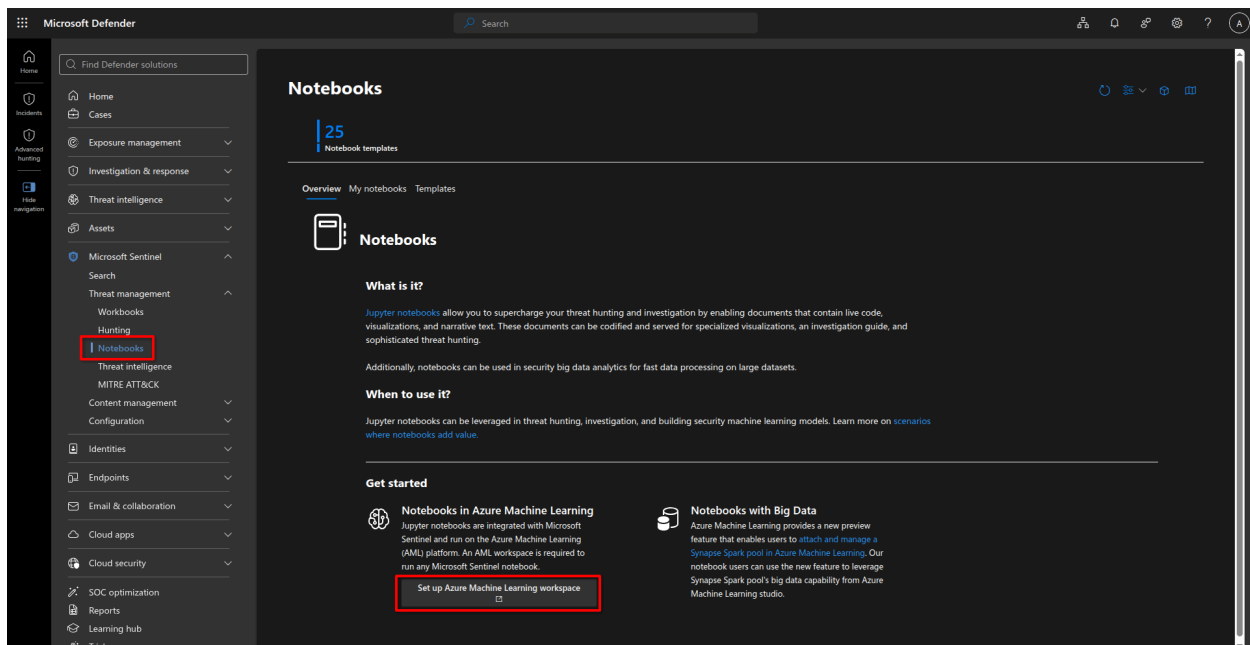
1. Left nav → **Microsoft Sentinel** → **Notebooks**

2. Launch a compute environment

Sentinel notebooks run inside **Azure Machine Learning (AML)** — you need a workspace and compute before opening anything.

1. From the Notebooks blade → click **Launch notebook environments** (or **Create Azure ML workspace** if none exists)
2. If prompted, create a new AML workspace:
 - Resource Group: **Sentinel-Lab-rg** (reuse your existing one)

- Workspace name: `sentinellab-aml`
 - Region: same as your Sentinel workspace
3. Once created, click **Launch notebook environments** again → this opens **Azure ML Studio** in a new tab
4. In AML Studio → **Compute** → **Compute instances** → **+ New**
- Name: `sentinel-notebook-vm`
 - Size: `STANDARD_DS3_V2` (or smallest available — this is a lab, not production)
 - Click **Create** and wait ~3–5 minutes for it to start



Microsoft Azure

Home > Azure Machine Learning

Azure Machine Learning

Create a machine learning workspace

Basics Inbound Access Outbound Access Encryption Identity Tags Review + create

Resource details

Every workspace must be assigned to an Azure subscription, which is where billing happens. You use resource groups like folders to organize and manage resources, including the workspace you're about to create. [Learn more about Azure resource groups.](#)

Subscription  Ayman-Visual Studio Enterprise Subscription - MPN

Resource group  Sentinel-Lab-rg
[Create new](#)

Workspace details

Configure your basic workspace settings like its storage connection, authentication, container, and more. [Learn more.](#)

Name  sentinellab-aml

Region  Sweden Central

Storage account  (new) sentinellabaml1054168415
[Create new](#)

Key vault  (new) sentinellabaml1005470480
[Create new](#)

Application insights  (new) sentinellabaml5590562263
[Create new](#)

Container registry  None
[Create new](#)

[Review + create](#) < Previous Next > Inbound Access

Microsoft Azure

Home > Azure Machine Learning

Azure Machine Learning

Create a machine learning workspace

 Validation passed

Basics Inbound Access Outbound Access Encryption Identity Tags Review + create

Subscription	Ayman-Visual Studio Enterprise Subscription - MPN
Resource group	Sentinel-Lab-rg
Region	Sweden Central
Name	sentinellab-aml
Storage account	(new) sentinellabaml1054168415
Key vault	(new) sentinellabaml1005470480
Application insights	(new) sentinellabaml5590562263
Container registry	None
Connectivity method	Enable public access from all networks
Network isolation	Public
Encryption type	Microsoft-managed keys
Identity type	System assigned
Enable HBI Flag	Disabled
Storage account access type	Credential-based
Shared key access	Enabled

[Create](#) < Previous Next > [Download a template for automation](#)

Microsoft Azure

Home

sentinellab-aml
Azure Machine Learning workspace

Search resources, services, and docs (G+I)

Copilot

aymani@globalbrandsgr...
GLOBAL BRANDS GROUP (GLO...

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Events

Settings

Monitoring

Automation

Support + troubleshooting

Essentials

Resource group : [Sentinel-Lab-rg](#)

Location : Sweden Central

Subscription : [Ayman-Visual Studio Enterprise Subscription - MPN](#)

Storage : [sentinellabaml1054168415](#)

Provisioning State : Succeeded

Studio web URL : <https://ml.azure.com?tid=c75c9052-d540-4fa5-914d-bf02683d7575&wsid=/subscriptions/...>

Container Registry : ...

Key Vault : [sentinellabaml1005470480](#)

Application Insights : [sentinellabaml5590562263](#)

MLflow tracking URI : [azureml://swedencentral.api.azureml.ms/mlflow/v1.0/subscriptions/60ba34d6-e952-486b-...](#)

JSON View

Work with your models in Azure Machine Learning Studio

The Azure Machine Learning Studio is a web app where you can build, train, test, and deploy ML models. Launch it now to start exploring, or learn more about the Azure Machine Learning studio.

[Launch studio](#)

Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Foundry | Azure Machine Learning

global brands group > sentinellab-aml > Notebooks

Notebooks

Files Samples

Users

ayman

Authoring

Notebooks

Automated ML

Designer

Prompt flow

Assets

Data

Jobs

Components

Pipelines

Environments

Models

Endpoints

Manage

Compute

Monitoring

Data Labeling

Connections

Notebooks is your space to add, browse, and edit files.

You can add files of any type, including Jupyter Notebooks (.ipynb). The files you see here are stored in the workspace file share, and are accessible and shared within the workspace.

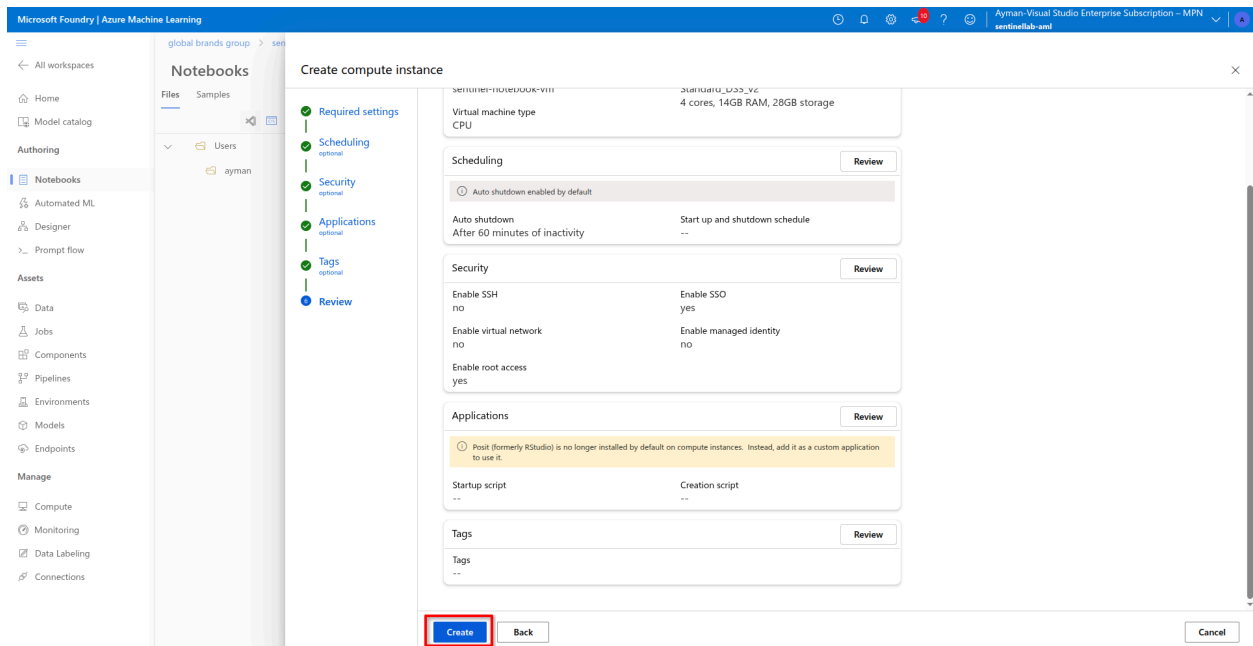
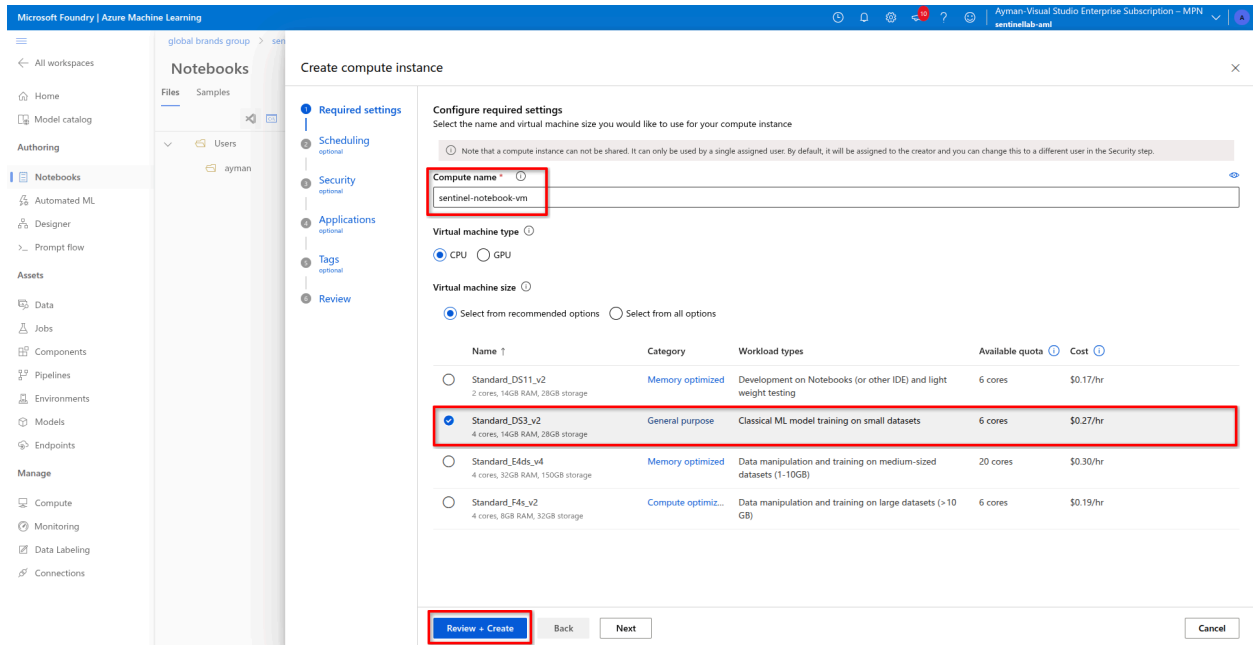
In order to run notebooks and scripts, you must connect to an Azure Machine Learning compute resource. Once a notebook or terminal is connected, you can access all workspace assets including experiment details, data, models, and more. [Learn more](#)

[+ Files](#) [Create compute](#)

[View Azure Machine Learning tutorials](#)

[View Release Notes to learn more about the latest features](#)

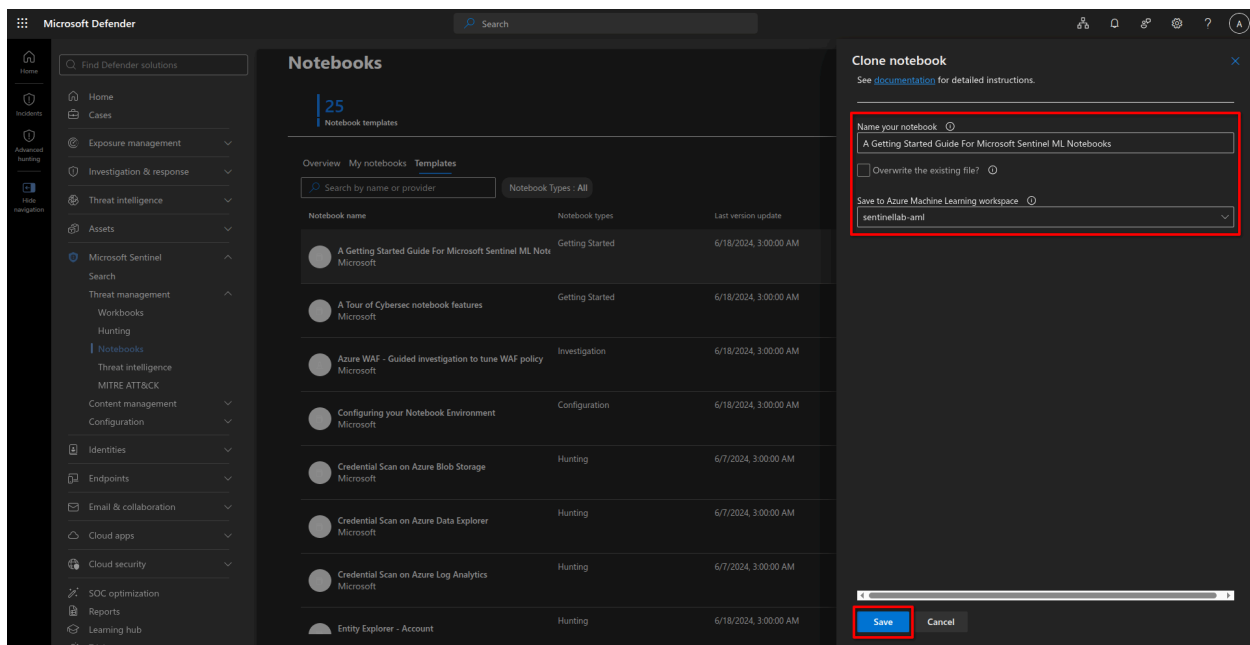
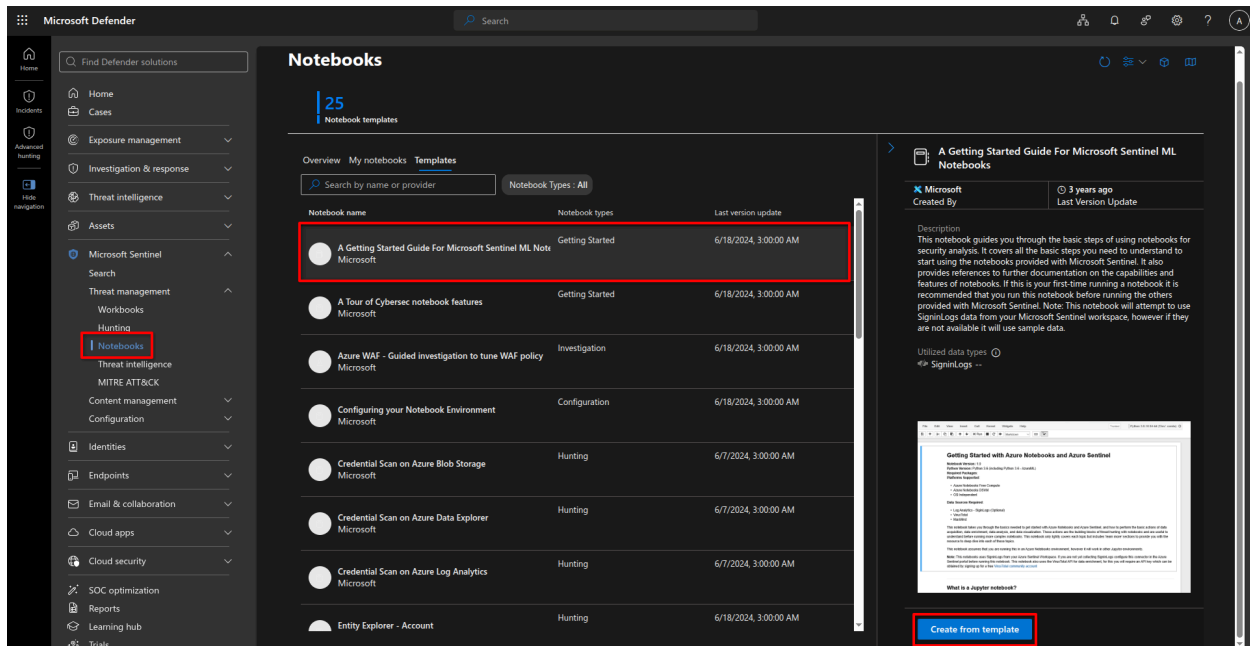
[Notebooks documentation](#)

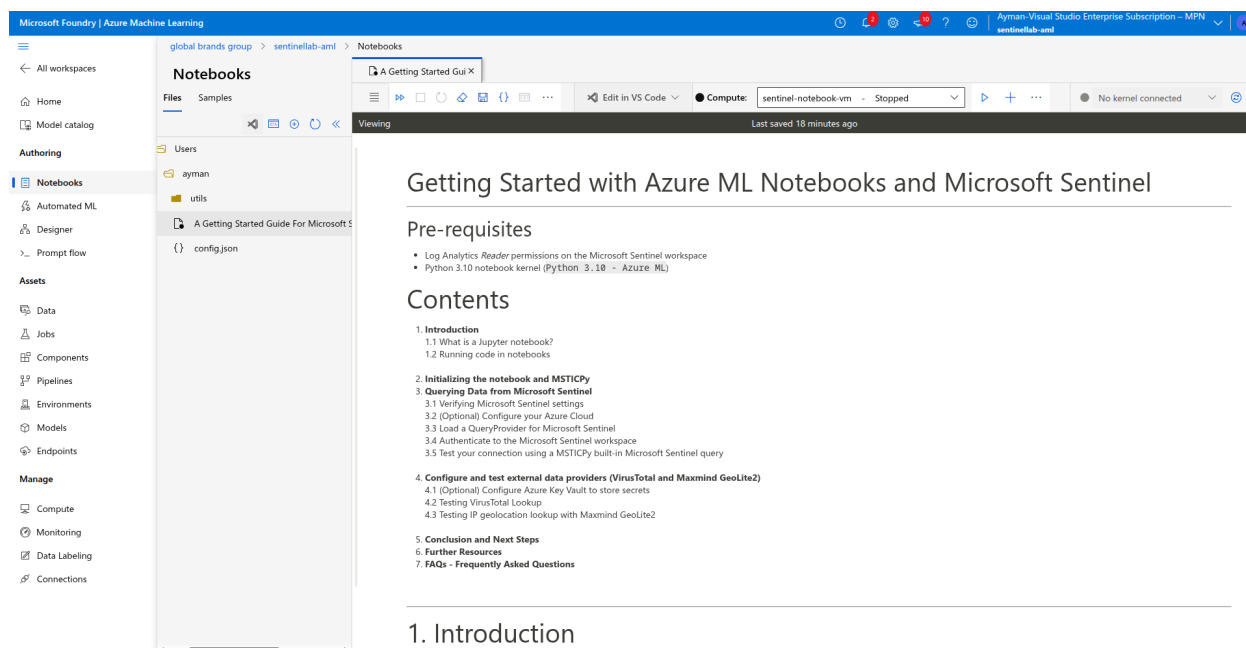
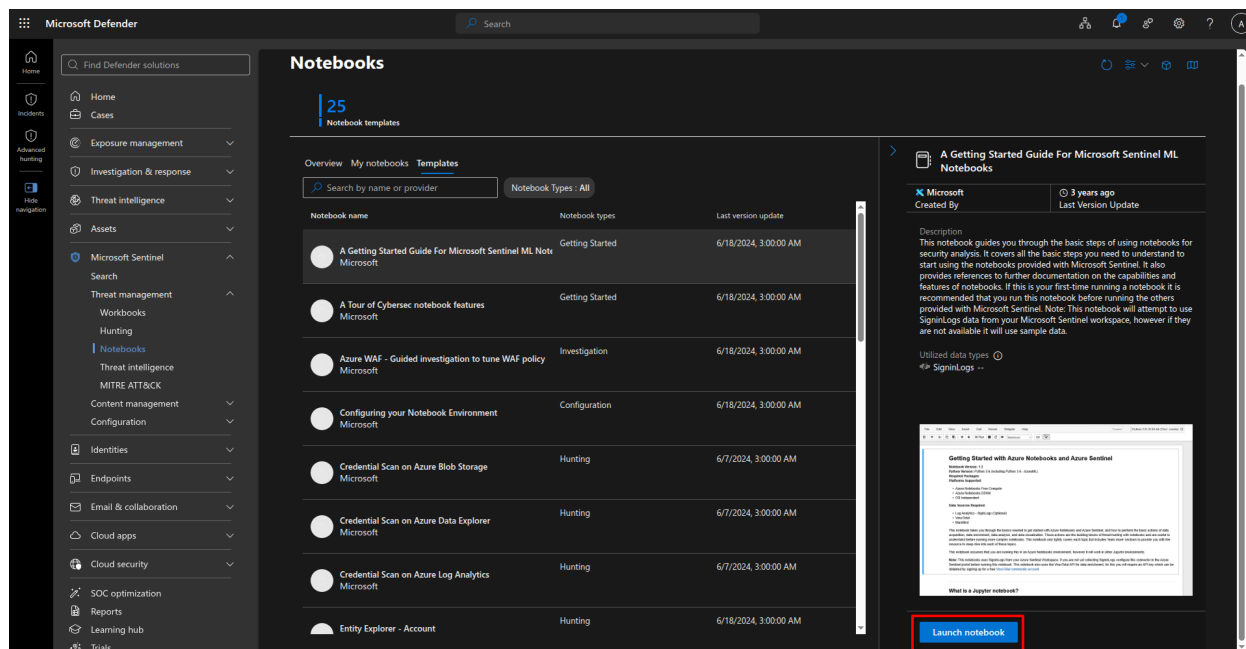


3. Clone the MSTICPy sample notebooks

1. Back in Sentinel → **Notebooks** → **Templates** tab
2. Pick a sample, e.g. "**A Getting Started Guide For Microsoft Sentinel ML Notebooks**" or "**Guided Hunting Notebook**"
3. Click **Save notebook** (this clones it into your AML workspace's file storage, usually under a **Users/<your-alias>** folder)

4. Click **Go to notebook** → opens the AML Studio Notebooks editor with the **.ipynb** file loaded





4. First-run setup inside the notebook

The first cell in almost every MSTICPy sample notebook handles package install and config check. Run cells top-to-bottom with **Shift+Enter**.

1. **Cell 1 — install/upgrade MSTICPy** (may take a couple of minutes on first run):

```
!pip install --upgrade msticpy[azure]
```

2. Cell 2 — imports and config check:

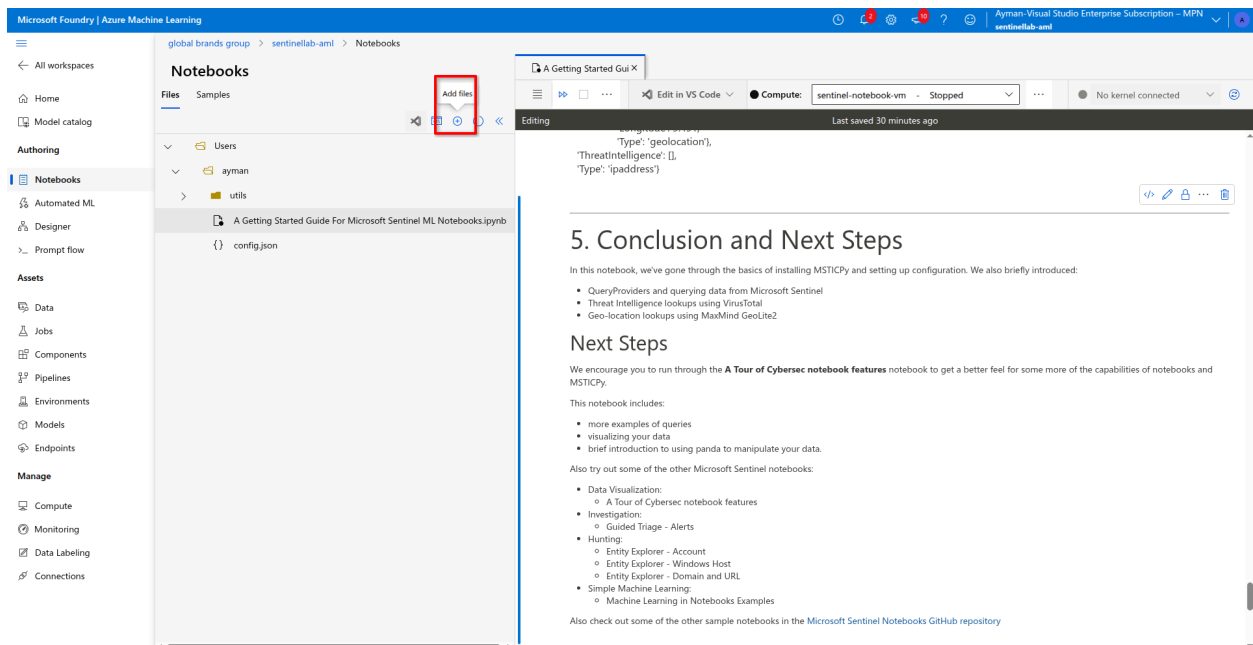
```
# import some modules needed in this cell
from IPython.display import display, HTML

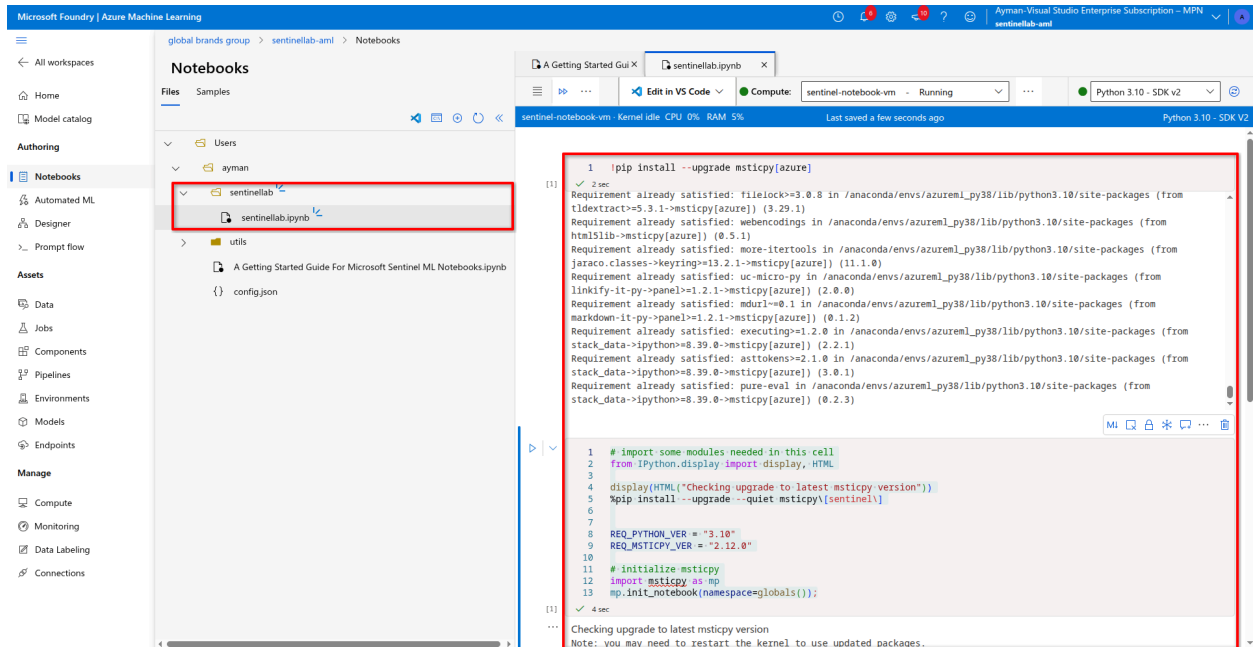
display(HTML("Checking upgrade to latest msticpy version"))
%pip install --upgrade --quiet msticpy\[sentinel\]

REQ_PYTHON_VER = "3.10"
REQ_MSTICPY_VER = "2.12.0"

# initialize msticpy
import msticpy as mp
mp.init_notebook(namespace=globals());
```

This checks for a `msticpyconfig.yaml` file and will prompt you if one isn't found.





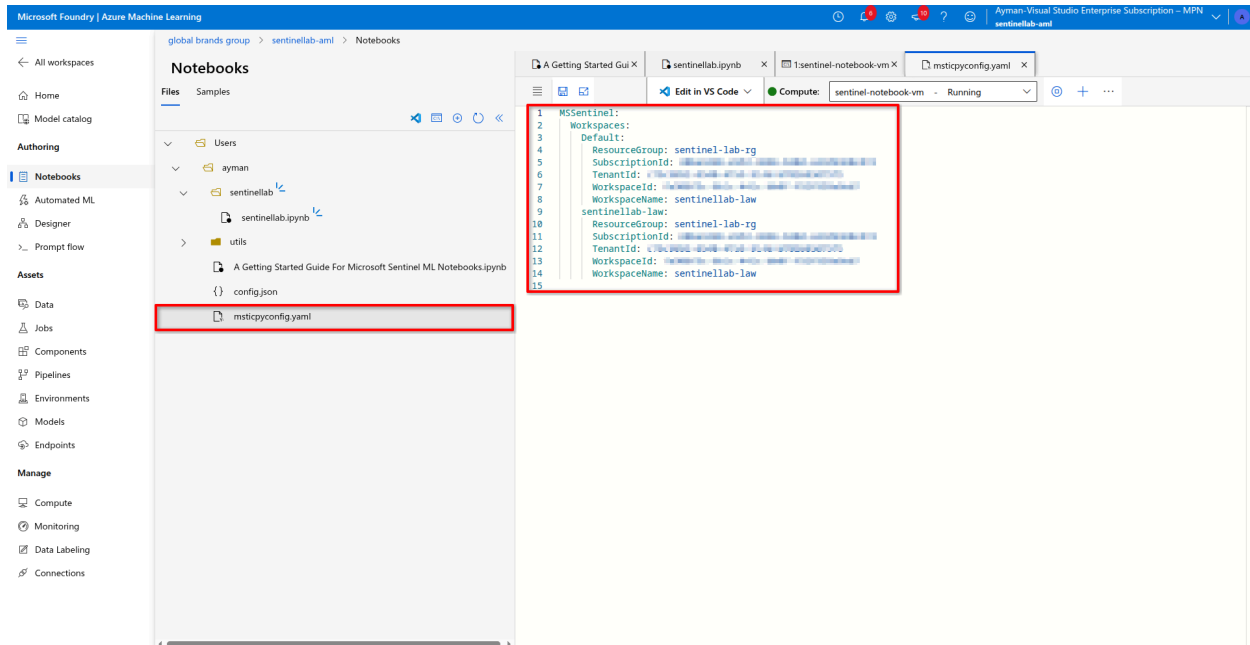
5. Configure `msticpyconfig.yaml` (workspace connection)

MSTICPy needs to know which Log Analytics/Sentinel workspace to query.

1. In the AML Studio file browser (left panel), check if `msticpyconfig.yaml` already exists at the root — Sentinel-generated notebooks often auto-create one pointing at your workspace
2. If not, create it (New File → `msticpyconfig.yaml`) with:

```
AzureSentinel:
  Workspaces:
    Default:
      WorkspaceId: "<your-log-analytics-workspace-id>"
      TenantId: "<your-azure-ad-tenant-id>"
```

- **WorkspaceId:** Sentinel → **Settings** → **Workspace settings** → copy **Workspace ID**
 - **TenantId:** Entra ID → **Overview** → copy **Tenant ID**
3. Point MSTICPy at the config file (usually auto-detected if it's in the same folder, otherwise set an env var):



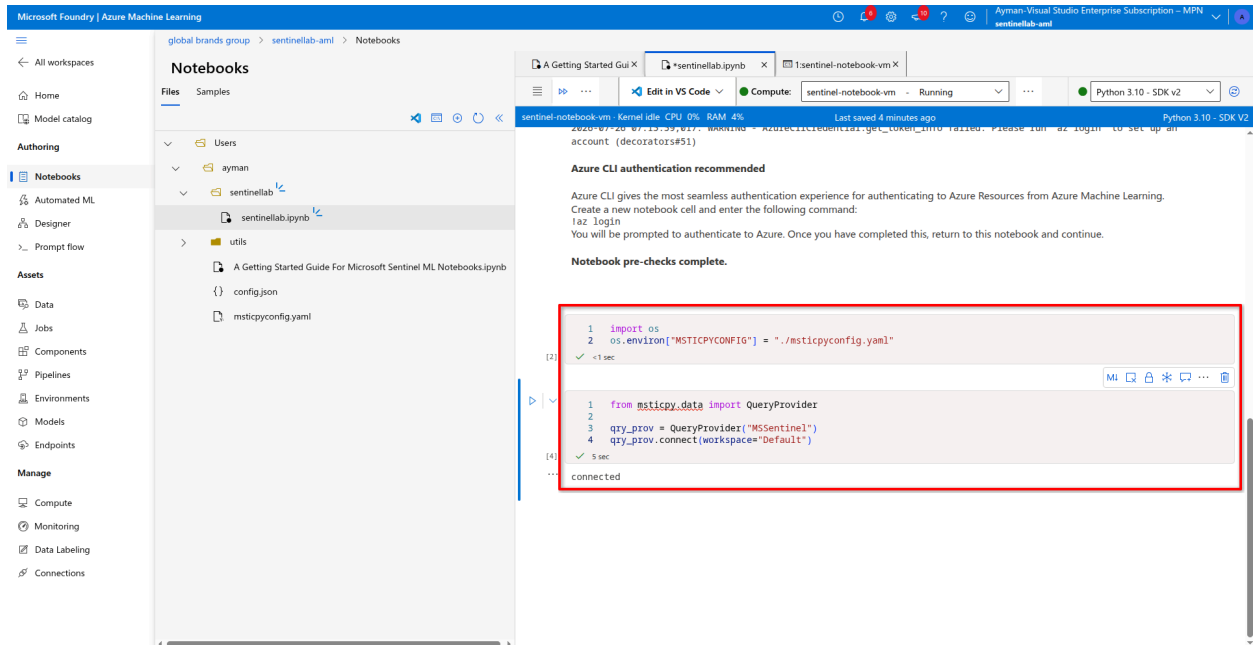
```
import os
os.environ["MSTICPYCONFIG"] = "./msticpyconfig.yaml"
```

6. Authenticate and connect to the workspace

```
from msticpy.data import QueryProvider

qry_prov = QueryProvider("MSSentinel")
qry_prov.connect(workspace="Default")
```

- This triggers an **interactive device-code login** — a popup/output will show a URL (microsoft.com/devicelogin) and a code
- Open that URL in another tab, enter the code, sign in with the same account you use for the Sentinel tenant
- Once authenticated, `qry_prov` is live and connected to your workspace



7. Run a pull against your lab data

Test the connection against the RDP brute-force telemetry you generated in Section C/D:

```
from datetime import datetime, timedelta

end = datetime.utcnow()
start = end - timedelta(days=1)

df = qry_prov.WindowsSecurity.list_host_logons(
    start=start,
    end=end,
    add_query_items="| where EventID == 4625"
)
df.head(20)
```

If `list_host_logons` isn't available in your provider version, fall back to raw KQL:

```
df = qry_prov.exec_query("""
SecurityEvent
| where EventID == 4625
```



```

| project TimeGenerated, Account, IpAddress, Computer
| take 20
"""
df

```

The screenshot shows the Microsoft Foundry | Azure Machine Learning interface. On the left, there's a sidebar with navigation options like 'All workspaces', 'Home', 'Model catalog', 'Authoring', 'Assets', 'Manage', etc. The main area displays a Jupyter Notebook titled 'sentinel-lab.ipynb'. The notebook code is as follows:

```

1 df = qry_prov.exec_query("""
2   SecurityEvent
3   | where EventID == 4625
4   | project TimeGenerated, Account, IpAddress, Computer
5   | take 20
6   """)
7 df

```

The output of the notebook is a table with 17 rows and 4 columns: TimeGenerated, Account, IpAddress, and Computer. The data is as follows:

	TimeGenerated	Account	IpAddress	Computer
0	2026-07-20 11:55:27.737065+00:00	-IMohamd	41.35.46.155	winServer2019
1	2026-07-20 11:55:45.415592+00:00	-IMohamd	41.35.46.155	winServer2019
2	2026-07-20 11:57:07.051307+00:00	-IMohamd	156.215.142.67	winServer2019
3	2026-07-20 11:57:22.432031+00:00	-IMohamd	156.215.142.67	winServer2019
4	2026-07-20 12:00:55.496317+00:00	-IMohamd	156.215.142.67	winServer2019
5	2026-07-20 12:01:05.120513+00:00	-IMohamd	156.215.142.67	winServer2019
6	2026-07-20 12:01:16.147779+00:00	-IMohamd	156.215.142.67	winServer2019
7	2026-07-20 12:02:22.780258+00:00	-IMohamd	41.35.46.155	winServer2019
8	2026-07-20 12:02:35.357320+00:00	-IMohamd	41.35.46.155	winServer2019
9	2026-07-20 12:19:56.310476+00:00	-IMohamd	41.35.46.155	winServer2019
10	2026-07-20 12:20:05.044563+00:00	-IMohamd	41.35.46.155	winServer2019
11	2026-07-20 12:20:20.212330+00:00	-IMohamd	41.35.46.155	winServer2019
12	2026-07-20 12:20:29.515414+00:00	-IMohamd	41.35.46.155	winServer2019
13	2026-07-20 12:20:37.703926+00:00	-IMohamd	41.35.46.155	winServer2019
14	2026-07-20 12:27:50.067356+00:00	-IMohamd	41.35.46.155	winServer2019
15	2026-07-20 12:27:59.880945+00:00	-IMohamd	41.35.46.155	winServer2019
16	2026-07-20 12:28:11.192266+00:00	-IMohamd	41.35.46.155	winServer2019
17	2026-07-20 12:36:52.831544+00:00	-IMohamd	41.35.46.155	winServer2019

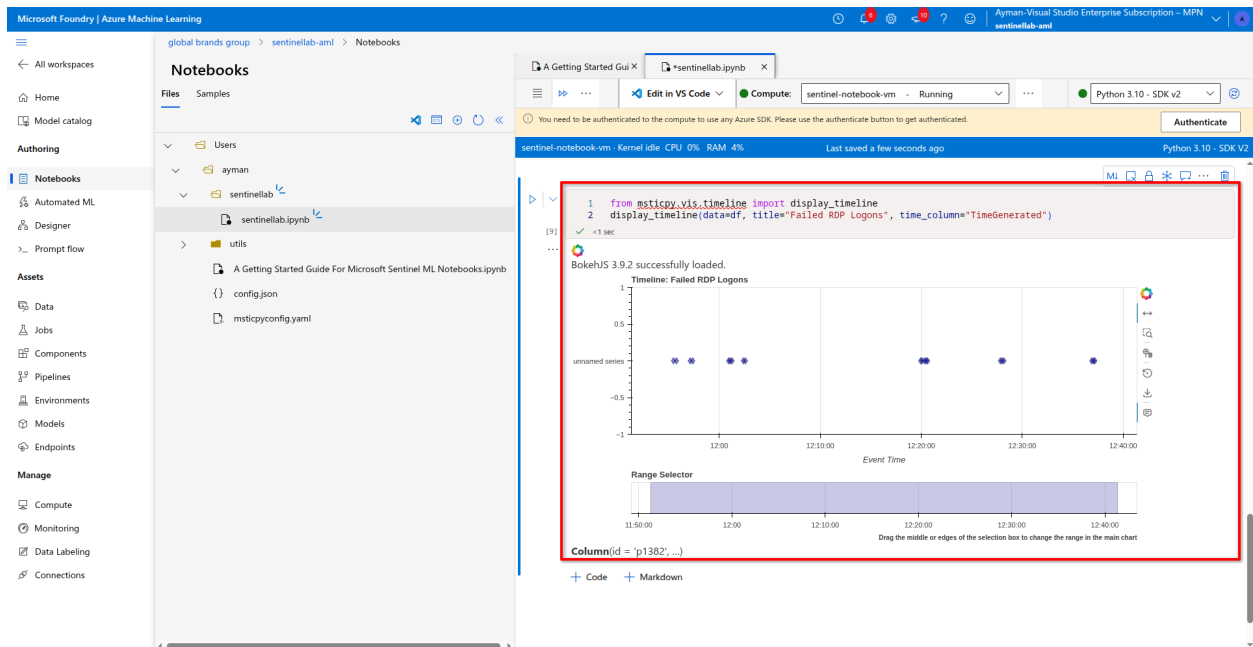
8. Visualize and pivot (typical use cases)

1. Timeline of failed logons:

```

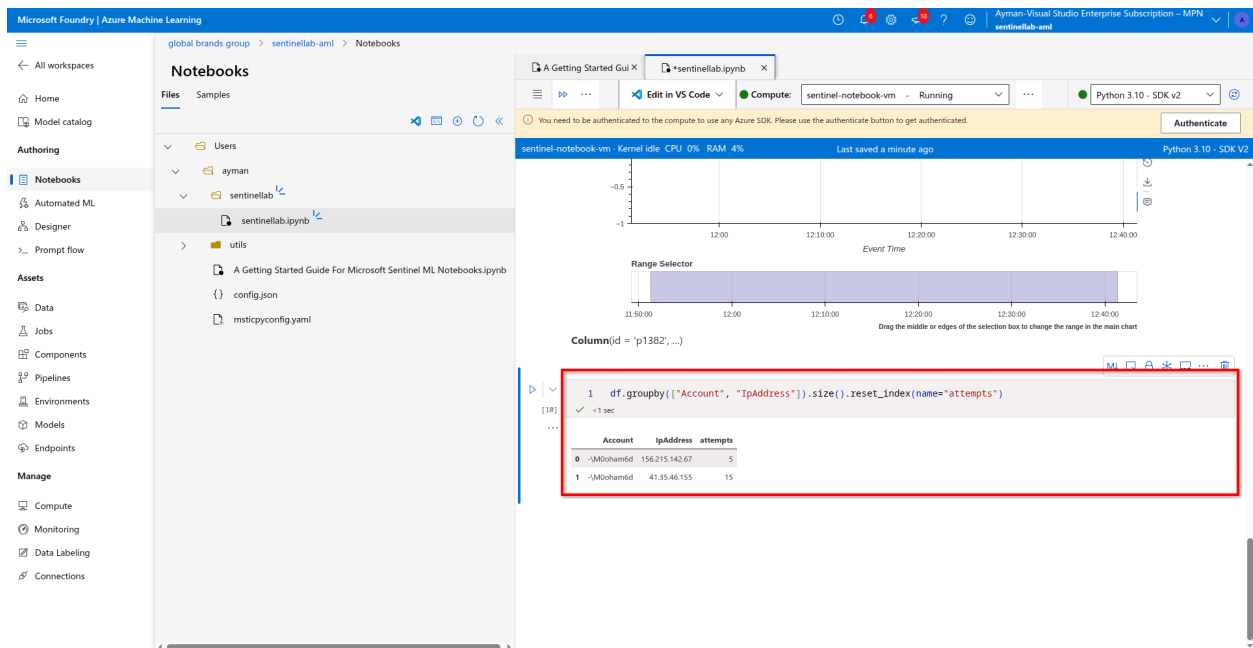
from msticpy.vis.timeline import display_timeline
display_timeline(data=df, title="Failed RDP Logons", time_column="TimeGenerated")

```



2. Investigation graph equivalent — relate accounts to source IPs:

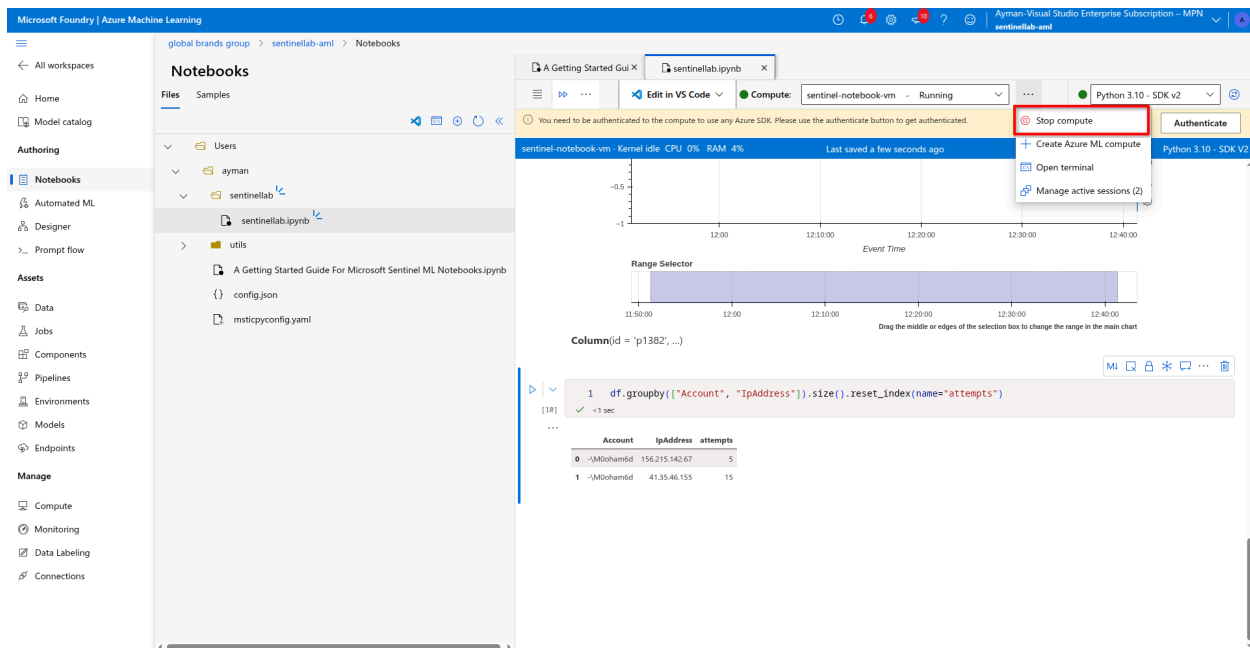
```
df.groupby(["Account", "IpAddress"]).size().reset_index(name="attempts")
```



9. Save and shut down compute (avoid ongoing cost)

AML compute instances bill by the hour while running, even if idle — this is the easy-to-forget cost leak in this section.

1. File → **Save** (or Ctrl+S) to persist notebook changes back to the workspace storage
2. AML Studio → **Compute** → **Compute instances** → select **sentinel-notebook-vm** → **Stop**
3. Confirm status changes to **Stopped** before moving on



I. Investigating incidents & long-range searches

1. Use the incident Graph to drill into entities and connections.
2. Open an entity (user/device/IP) from an incident to see the entity page with timeline and related alerts.
3. Search long time spans: Logs → set **Time range** to custom (e.g., last 90 days) and run queries (respect retention limits).
4. Restoring archived logs: follow Log Analytics restore steps in the portal if you moved data to archive.

J. End-to-end testing flow (example)

1. Post 10 failed logon entries for `testuser` from `10.0.0.5` into `CustomSecurityEvents_CL` via Data Collector API.
 2. Confirm ingestion: `CustomSecurityEvents_CL | where Account == "testuser" | take 20`
 3. Ensure analytics rule thresholds match the test data (≥ 5 or ≥ 10 as configured).
 4. Wait for rule execution and confirm incident creation.
 5. Open incident → create Case → assign owner → run playbook and confirm Teams message and comment.
-

K. Cleanup checklist

1. Delete test VMs: Portal → Virtual machines → select VM → Delete.
2. Delete Logic Apps playbooks: Portal → Logic apps → select playbook → Delete.
3. Delete Sentinel / Log Analytics workspace: disable Sentinel or delete the workspace resource.
4. Delete resource group: Portal → Resource groups → select `Sentinel-Lab-rg` → Delete resource group (removes most created resources).
5. Verify Cost Management for any remaining running resources.